

IMPERIAL

APPLIED MATHEMATICS RESEARCH PROJECT

Bayesian nonparametric inference in Vlasov-Fokker-Planck models

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The work contained in this thesis is my own work unless otherwise stated.

SUMMARY

We study the problem of recovering the interaction potential W in the Vlasov–Fokker–Planck equation from discrete noisy observations of the phase-space density or of its spatial marginal, adapting the Bayesian nonparametric framework recently developed for the overdamped McKean–Vlasov model to the kinetic setting. Both measurement schemes are cast as random-design nonlinear regression problems, and inference is performed by placing a rescaled Gaussian process prior on the class of symmetric, mean-zero potentials. Conditionally on a well-posedness and regularity hypothesis, and on an entropy stability estimate for the equation, we show that both solution maps are forward regular with smoothing index governed by the regularity of the initial state, and we derive posterior contraction theorems for the forward quantities, at a rate δ_N which features the torus dimension d only and where N is the number of measurements. The theory is complemented by numerical experiments based on a structure-preserving Hermite–Fourier spectral solver, in which the hypocoercive relaxation mechanism is directly visible at the level of the discretised generator, coupled with function-space Monte Carlo. The simulations empirically confirm the conditional Lipschitz stability of the forward maps, exhibit accurate recovery of the dynamically excited Fourier modes of W , show that higher frequencies remain as uncertain as the prior and that density-only observations prove as informative as full phase-space observations.

All experiments are reproducible from the companion notebook available [here](#).

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INTRODUCTION

We study an adaptation of [1] to the problem of conducting statistical inference in the class of kinetic dynamical systems for which the state at time $t \in [0, T]$ is described by the phase space probability density $f = f_W(t, q, p)$ solving the nonlinear Vlasov-Fokker-Planck equation

$$\begin{aligned} \frac{\partial f}{\partial t} + p \cdot \nabla_q f - \nabla_q (V + W * \rho) \cdot \nabla_p f &= \gamma \nabla_p \cdot (pf + \beta^{-1} \nabla_p f), \\ f(0, \cdot, \cdot) &= \phi, \end{aligned} \tag{VFP}$$

where $q \in \mathbb{T}^d$ and $p \in \mathbb{R}^d$ denote the position and momentum variables, respectively, while ϕ is the probability density of the initial phase-space state, γ denotes a friction coefficient and β the inverse temperature, V is a known confining potential and W is an unknown symmetric interaction potential with Lipschitz gradient, which is to be recovered from a discrete statistical measurement of the evolution of either the full phase space state f of the system over time, or of its macroscopic spatial density, which is given by

$$\rho = \rho_W(t, q) = \int_{\mathbb{R}^d} f_W(t, q, p) \, dp$$

and makes the equation into a nonlocal-in-space and nonlinear problem, falling in the framework of [2].

This kinetic partial differential equation (PDE) was studied, in the absence of an interaction potential, first by Klein and later by Kramers in [3] as the Fokker-Planck equation governing the evolution of an underdamped Brownian particle subject to friction and stochastic forcing. Vlasov introduced the nonlinear and nonlocal term in [4], to model a system of charged particles for which long-range collective interactions dominate over binary collisions.

Specifically, the equation arises as the mean-field limit, as $n \rightarrow \infty$, of an interacting particle system (IPS) in phase space with $n \in \mathbb{N}$ components, governed by the coupled system of stochastic differential equations (SDEs),

$$\begin{aligned} dq_t^i &= p_t^i \, dt, \quad i = 1, \dots, n \\ dp_t^i &= -\nabla_q V(q_t^i) \, dt - \frac{1}{n} \sum_{j \neq i} \nabla_q W(q_t^i - q_t^j) \, dt - \gamma p_t^i \, dt + \sqrt{2\gamma\beta^{-1}} \, dB_t^i, \end{aligned} \tag{IPS}$$

where the B_t^i are independent d -dimensional Brownian motions, and where the initial configuration $(q_0^i, p_0^i)_{i=1}^n$ is assumed to be exchangeable and chaotic: namely, the joint law of the particles' positions and momenta is invariant under permutations of the particles and, as n is taken to be large, the initial joint marginal $\phi_k^{(n)}$ for the first $k < n$ components factorizes:

$$\lim_{n \rightarrow \infty} \phi_k^{(n)}(q^1, p^1, \dots, q^k, p^k) = \prod_{j=1}^k \phi(q^j, p^j). \tag{Kac}$$

This notion was introduced in [5], under the name of ‘Boltzmann property’, referring to the physicist’s so-called *Stosszahlansatz* and, following early ideas of McKean in [6], the mean-field limit can be made rigorous by introducing the Langevin equation where the law of $(\bar{q}_t, \bar{p}_t)$ at each time $t \in [0, T]$ has density f solving (VFP). One can show that Kac’s chaos property (Kac) is propagated in time:

$$\lim_{n \rightarrow \infty} f_k^{(n)}(t, q^1, p^1, \dots, q^k, p^k) = \prod_{j=1}^k f(t, q^j, p^j),$$

where $f_k^{(n)}$ denotes the k -particle marginal density of the n -particle system; this *propagation of chaos* property, with quantitative rates, is proved in [7] by a synchronous coupling of (IPS) with n independent copies of (?), together with existence and uniqueness of solutions to the McKean SDE via a fixed-point contraction argument in Wasserstein distance on path space, in a general setting: see [8, 9, 10, 11] for (VFP).

To infer the interaction potential W , we adopt a Bayesian nonparametric framework, as in [2, 1]; see [12, 13] for related frequentist approaches to nonparametric inference in mean-field particle models, [14, 15, 16] for the Bayesian analysis of diffusions and [17, 18, 19] for the use of Gaussian process priors in such context. Since W enters (VFP) only through the convolved gradient $\nabla_q W * \rho$, it is at most identifiable up to an additive constant; we shall therefore impose the normalisation

$$\int_{\mathbb{T}^d} W(q) dq = 0$$

and regard W as a potentially infinite-dimensional parameter ranging over a class $\mathcal{W}$ of symmetric: $W(-q) = W(q)$, mean-zero potentials with Lipschitz gradient. We place a Gaussian process prior Π on W , supported on, or suitably projected onto, such class. The prior is subsequently updated to a posterior distribution using Bayes theorem, conditioned on discrete, noisy observations of the phase-space density or of its macroscopic spatial marginal. Throughout, we normalise the torus to have unit volume, write $\lambda_T = \text{unif}(0, T)$ for the uniform distribution on the time interval, and set $\lambda = \lambda_T \otimes dq = \text{unif}(\mathcal{X})$ on the time-space cylinder $\mathcal{X} = [0, T] \times \mathbb{T}^d$. Specifically, we consider two possible regression-type measurement models in which the observed quantity, corrupted by independent and identically distributed (i.i.d.) noise $\epsilon_i \sim_{\text{i.i.d.}} \mathcal{N}(0, \sigma_\epsilon^2)$ with known variance $\sigma_\epsilon^2 > 0$, independent of the design, is given by

- i. the macroscopic spatial density:

$$Y_i = \rho_W(t_i, q_i) + \epsilon_i,$$

with design points sampled discretely from the time-space cylinder, $(t_i, q_i) \sim_{\text{i.i.d.}} \lambda$;

- ii. the full state on the time-phase space $\mathcal{Z} = \mathcal{X} \times \mathbb{R}^d$:

$$Y_i = f_W(t_i, q_i, p_i) + \epsilon_i,$$

with $(t_i, q_i, p_i) \sim_{\text{i.i.d.}} \lambda \otimes \nu(p) dp$. Since no uniform probability measure exists on $\mathbb{R}^d$, a sampling density ν on the momentum variable must be chosen; we take ν to be a bounded and strictly positive probability density, for instance $\nu(p) \propto \langle p \rangle^{-2s}$ for some $s > d/2$, where $\langle p \rangle = (1 + |p|^2)^{1/2}$, or a Gaussian. The choice of ν fixes the design norm $L^2(\lambda \otimes \nu)$ in which recovery of f_W is assessed, and thereby balances the statistical weight assigned to the informative low-momentum region against the tails, where f_W decays rapidly; a Gaussian choice of ν is moreover naturally aligned with the Maxwellian-weighted L^2 spaces in which the kinetic Fokker–Planck dynamics is commonly analysed [20, 21]; cf. Remark 4 below.

As in the overdamped setting of [1], consistent recovery of W will additionally require a suitable non-degeneracy (deconvolvability) condition on the initial density ϕ , ruling out in particular initialisation at a stationary state of the dynamics; see Section 2 below. This methodology aligns with the broader field of Bayesian inverse problems for nonlinear PDEs as treated by [2], which we now review and that can be computationally resolved by combining function-space Markov Chain Monte Carlo (MCMC) algorithms with numerical PDE solvers, see [22, 23] and Section 3 for our implementation; polynomial-time computational guarantees for gradient-based MCMC in such settings are obtained in [24].

1. BAYESIAN NONLINEAR STATISTICAL INVERSE PROBLEMS

Before addressing the kinetic problem (VFP), we review the general framework of [2] for Bayesian inference in nonlinear PDE-constrained regression models, in the transparent model setting of an elliptic boundary value problem.

Let $\Omega \subset \mathbb{R}^d$ be a domain and consider a linear elliptic differential operator $\mathcal{L}_W$, indexed by some $W : \Omega \rightarrow \mathbb{R}$. Then, we look for a solution to the boundary value problem

$$\begin{cases} \mathcal{L}_W f = \phi, & \text{in } \Omega \\ f = \psi, & \text{on } \partial\Omega \end{cases} \quad (\text{BVP})$$

where ϕ is a smooth *source function* and ψ prescribes smooth boundary values.

The motivating problem is that of recovering W from solutions f_W to (BVP). To acquire approximate measurements of f_W from observations of the diffusion over a time interval $(0, T]$, we consider solutions

$$\{f = f_W(t, x) : x \in \Omega, t \in (0, T]\}$$

to the initial-boundary value problem

$$\begin{cases} \partial_t f = -\mathcal{L}_W f, & \text{in } \Omega \times (0, T] \\ f(t, \cdot) = 0, & \text{on } \partial\Omega \times (0, T] \\ f(0, \cdot) = \phi, & \text{on } \Omega \end{cases} \quad (\text{IBVP})$$

where ϕ is some specified initial condition. In fact, equipped with such Dirichlet boundary conditions, the elliptic operator has a spectral representation given by the Fourier decomposition

$$\mathcal{L}_W = \sum_{n \geq 1} \lambda_n (\nu_n, \cdot)_{L^2(\Omega)} \nu_n$$

in terms of the Hilbert eigenpairs $\{\lambda_n, \nu_n\}_n \subset (0, \infty) \times H_0^1(\Omega)$. Then, the unique solution to the diffusion equation is given by

$$f_W(t, \cdot)[\phi] = \sum_{n \geq 1} e^{-t\lambda_n} (\nu_n, \phi)_{L^2(\Omega)} \nu_n, \quad t \geq 0$$

and thus

$$\int_0^\infty f_W(t, \cdot)[\phi] dt = \sum_{n \geq 1} \lambda_n^{-1} (\nu_n, \phi)_{L^2(\Omega)} \nu_n = \mathcal{L}_W^{-1} \phi$$

provides the solution to (BVP) with $\psi \equiv 0$, whence the Riemann sum approximation $\sum_k f_W(t_k, \cdot)[t_k - t_{k-1}]$, where $f_W(t_k, \cdot)$ are measurements of the solution to (IBVP) at discrete equally spaced times $t_k \in (0, \infty)$, is the searched approximate measurement of steady state solutions.

We can also introduce a parameter space Θ consisting of a Borel-measurable subset of $L^2(\Omega)$, so that we can regard coefficients $W = W_\theta$ as being parametrised by $\theta \in \Theta$.

Finally, we define the *forward map* $\mathcal{G} : \theta \in \Theta \mapsto \mathcal{G}(\theta) = f_{W_\theta} \in L^2(\Omega)$.

The inverse problem is then that of recovering θ from ‘data’ $\mathcal{G}(\theta)$. In any realistic experiment, however, one does not observe the full function $\mathcal{G}(\theta)$: what is available is a finite collection of point evaluations, each contaminated by measurement error. In our setting, the solution f_W of (BVP) is sampled at design points $X_1, \dots, X_N \in \Omega$, and the recorded value at X_i differs from the true value $\mathcal{G}(\theta)[X_i]$ by a random perturbation ϵ_i , which we model as Gaussian.

The data set therefore consists of an i.i.d. sample $D_N = (Y_i, X_i)_{i=1}^N$ obeying the regression equation

$$Y_i = \mathcal{G}(\theta)[X_i] + \epsilon_i \quad \epsilon_i \sim \text{i.i.d. } \mathcal{N}(0, 1), \quad i = 1, \dots, N.$$

Writing P_θ for the distribution of a single observation pair (Y, X) , which admits the Lebesgue-design density

$$p_\theta(y, x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}|y - \mathcal{G}(\theta)[x]|^2\right),$$

the full sample is governed by the N -fold product measure P_θ^N on $(\mathbb{R} \times \Omega)^N$.

We shall call a probability measure Π on Θ the *prior*, such that the *posterior distribution* is given by

$$d\Pi(\theta \mid D_N) = \frac{\prod_{i=1}^N p_\theta(Y_i, X_i) d\Pi(\theta)}{\int_{\Theta} \prod_{i=1}^N p_\theta(Y_i, X_i) d\Pi(\theta)} = \frac{e^{\ell_N(\theta)} d\Pi(\theta)}{\int_{\Theta} e^{\ell_N(\theta)} d\Pi(\theta)} \quad (\Pi)$$

where the log-likelihood of the data

$$\ell_N(\theta) = \sum_{i \leq N} \ell_i(\theta), \quad \ell_i(\theta) = -\frac{1}{2} |Y_i - \mathcal{G}(\theta)[X_i]|^2.$$

Since $\mathcal{G}$ is nonlinear, the posterior above rarely admits a tractable closed form, and in practice one samples from it by constructing a Markov chain whose invariant distribution coincides with the posterior. Two features make this computationally viable: each step of the chain demands only an evaluation of $\ell_N(\theta)$, hence of the forward operator $\mathcal{G}$ and never of the inversion map $\mathcal{G}^{-1}$, and the normalising constant in the denominator cancels in the sampling scheme and need not be computed; see Section 1.2.

Nonlinearity also restricts which priors can support valid inference on the regression function $\mathcal{G}(\theta)$. The benchmark for success is asymptotic: as the sample size N grows to infinity, the posterior should place all of its mass in vanishing neighbourhoods of the data-generating ‘ground truth’ parameter θ_0 , degenerating in the limit to the point mass δ_{θ_0} . This property goes under the name of *posterior consistency*.

1.1. Gaussian process priors. We take Π to be the Borel probability measure induced on a closed linear subspace Θ of the Hilbert space $L^2(\Omega)$ by a Gaussian random field, so that Π is Gaussian in the infinite-dimensional sense.

A normed subspace $\mathcal{R} \subseteq \Theta$ carrying the full mass of Π , i.e. such that

$$\Pi(\mathcal{R}) = \Pi(\theta \in \Theta : \|\theta\|_{\mathcal{R}} < \infty) = 1,$$

will play the role of a ‘regularisation space’ for the problem. The prototypical example to keep in mind is $\mathcal{R} = H^\beta(\Omega)$, paired with the reproducing kernel Hilbert space (RKHS) $\mathcal{H} = H^\alpha(\Omega)$ for some $\alpha > \beta + d/2$; see [25, 26, 27, 28].

A subtlety arises at this point: because $\mathcal{G}$ is nonlinear, the distribution it induces on the regression functions $\mathcal{G}(\theta)$ is no longer Gaussian, and it is not obvious that a Gaussian prior on θ produces a sensible prior at the level of $\mathcal{G}(\theta)$. This difficulty is resolved whenever $\mathcal{G}$ enjoys a Lipschitz-type continuity property, in appropriate norms, over a sufficiently large region of the parameter space Θ .

DEFINITION 1.1. We say that the map $\mathcal{G} : \Theta \rightarrow L^2(\Omega)$ is *forward regular* with *smoothing index* $\kappa \geq 0$ if, for every radius $M > 0$, one can find constants $U = U(M) \geq 1$ and $L = L(M) > 0$ for which it holds

i) *uniform boundedness*:

$$\sup_{\theta \in \Theta \cap B_{\mathcal{R}}(M)} \sup_{x \in \Omega} |\mathcal{G}(\theta)[x]| \leq U;$$

ii) *Lipschitz continuity*:

$$\|\mathcal{G}(\theta) - \mathcal{G}(\bar{\theta})\|_{L^2(\Omega)} \leq L \|\theta - \bar{\theta}\|_{H^\kappa(\Omega)^*}$$

for all $\theta, \bar{\theta} \in \Theta \cap B_{\mathcal{R}}(M)$.

The index $\kappa \geq 0$ in Definition 1.1 quantifies how strongly $\mathcal{G}$ regularises its input: larger values correspond to forward maps that damp irregular perturbations of θ more aggressively, since the Lipschitz estimate is then taken in a weaker (negative-order) norm.

We build the prior $\Pi = \Pi_N$ from a fixed base prior $\bar{\Pi}$ by shrinking its draws:

$$\theta = N^{-d/(4\alpha+4\kappa+2d)} \bar{\theta}, \quad \bar{\theta} \sim \bar{\Pi}$$

where $\alpha > 0$ is a ‘regularity’ parameter to be chosen and $\kappa \geq 0$ is the *forward smoothing degree* of $\mathcal{G}$. The multiplicative factor $N^{-d/(4\alpha+4\kappa+2d)}$ contracts draws from the base prior towards the origin, and this contraction acts as an additional, sample-size-dependent regularisation of the posterior. In the finite-dimensional caricature where $\mathcal{H}$ has finite dimension, the rescaled prior possesses a density proportional to

$$d\Pi(\cdot) \propto \exp\left(-\frac{1}{2}N^{d/(2\alpha+2\kappa+d)}\|\cdot\|_{\mathcal{H}}^2\right)$$

so that the penalty attached to the RKHS norm $\|\theta\|_{\mathcal{H}}$ is inflated by a power of N .

Let us introduce the (semi-)metric

$$d_{\mathcal{G}}(\theta, \bar{\theta}) = \|\mathcal{G}(\theta) - \mathcal{G}(\bar{\theta})\|_{L^2(\Omega)}$$

on Θ . Finally, given $\bar{\Theta} \subseteq \Theta$ and $\eta > 0$, let $N(\bar{\Theta}, d_{\mathcal{G}}, \eta)$ stand for the η -covering number of $\bar{\Theta}$ with respect to $d_{\mathcal{G}}$: the smallest cardinality of a collection of $d_{\mathcal{G}}$ -balls of radius η whose union contains $\bar{\Theta}$.

THEOREM 1.2 (POSTERIOR CONTRACTION). Let $\mathcal{G} : \Theta \rightarrow L^2(\Omega)$ be forward regular with smoothing index $\kappa \geq 0$ and regularisation space $\mathcal{R}$, and let $\bar{\Pi}$ be a centred Gaussian Borel probability measure on Θ with $\bar{\Pi}(\mathcal{R}) = 1$, whose RKHS $\mathcal{H}$ satisfies, for some $\alpha > 0$, the continuous embedding

$$\mathcal{H} \hookrightarrow H^{\alpha}(\Omega) \quad \text{if } \kappa < 1/2 \quad \text{or} \quad \mathcal{H} \hookrightarrow H_0^{\alpha}(\Omega) \quad \text{if } \kappa \geq 1/2.$$

Define the contraction rate

$$\delta_N \equiv N^{-(\alpha+\kappa)/(2\alpha+2\kappa+d)},$$

so that $N\delta_N^2 \rightarrow \infty$, and let Π_N be the rescaled prior: $\Pi_N = \text{Law}(N^{-d/(4\alpha+4\kappa+2d)}\bar{\theta})$, $\bar{\theta} \sim \bar{\Pi}$. Assume the ground truth satisfies $\theta_0 \in \mathcal{H} \cap \mathcal{R}$, and set $U \equiv U(\|\theta_0\|_{\mathcal{R}} \vee 1) \geq \|\mathcal{G}(\theta_0)\|_{\infty}$, the *envelope constant* furnished by forward regularity.

Then, the sets

$$\mathcal{B}_N \equiv \{\theta \in \Theta : d_{\mathcal{G}}(\theta, \theta_0) \leq \delta_N, \|\mathcal{G}(\theta)\|_{\infty} \leq U\}$$

satisfy, for all N large enough,

$$\Pi_N(\mathcal{B}_N) \geq e^{-AN\delta_N^2} \quad \text{for some } A > 0$$

and there moreover exists a sequence of Borel sets $\Theta_N \subset \Theta$ for which

$$\Pi_N(\Theta_N^c) \leq e^{-BN\delta_N^2} \quad \text{for some } B > A + 2,$$

and such that, for all $\bar{m} > 0$ large enough,

$$\log N(\Theta_N, d_{\mathcal{G}}, \bar{m}\delta_N) \leq N\delta_N^2.$$

In particular, ‘regularisation sets’ can be chosen as

$$\Theta_N = \{\theta \in \mathcal{R} : \theta = \theta_1 + \theta_2, \|\theta_1\|_{H^{-\kappa}} \leq M\delta_N, \|\theta_2\|_{\mathcal{H}} \leq M, \|\theta\|_{\mathcal{R}} \leq M\}$$

for $M = M(B)$ sufficiently large and, for all $0 < b < B - A - 2$, we can choose $L = L(A, B, \bar{m}, b)$ large enough such that

$$P_{\theta_0}^N \left\{ \Pi_N \left(\theta \in \Theta_N, d_{\mathcal{G}}(\theta, \theta_0) \leq L\delta_N C_U^{-1} \mid D_N \right) \leq 1 - e^{-bN\delta_N^2} \right\} \rightarrow 0$$

as $N \rightarrow \infty$, where $C_U \equiv (1 - e^{-U^2/2})/2U^2$.

REMARK 1. An inspection of the proof of [2, Theorem 2.2.2] shows that the statement of Theorem 1.2 holds, with identical proof, in the following generality: the design points are drawn i.i.d. from an arbitrary fixed Borel probability measure μ on a measurable design space $\mathcal{X}_{\text{des}}$, the forward map takes values in $L^2(\mathcal{X}_{\text{des}}, \mu)$, and forward regularity (Definition 1.1) is understood with $L^2(\mathcal{X}_{\text{des}}, \mu)$ replacing $L^2(\Omega)$ and with the parameter domain Ω replaced by any bounded manifold, in particular $\mathbb{T}^d$. Indeed, the argument uses the design only through the i.i.d. structure and the uniform boundedness of the regression functions $\{\mathcal{G}(\theta) : \theta \in \Theta \cap B_{\mathcal{R}}(M)\}$, which enter via

the concentration and testing bounds in the design norm; no boundedness or uniformity of the design measure is required. Moreover, on the boundaryless manifold $\mathbb{T}^d$, the dual space $(H^\kappa(\mathbb{T}^d))^*$ coincides with $H^{-\kappa}(\mathbb{T}^d)$ and the distinction between H^α and H_0^α in the embedding hypothesis is vacuous, its role being taken over by the closed mean-zero and symmetry constraints built into $\mathcal{H}$.

At this point, there is hope that a Bayes method, successfully inferring the regression maps $\mathcal{G}(\theta)$, also recovers the parameter θ of interest in the inverse problem if the forward map $\mathcal{G}$ satisfies a ‘stability estimate’ of the following kind:

$$\|\theta_1 - \theta_2\| \lesssim \|\mathcal{G}(\theta_1) - \mathcal{G}(\theta_2)\|^\eta, \quad \theta_1, \theta_2 \in \bar{\Theta}$$

for a suitable norm, sufficiently large subsets $\bar{\Theta} \subset \Theta$ and ‘stability’ Hölder exponent $\eta > 0$. These can be regarded as a ‘quantitative statement’ about the injectivity of the forward map $\theta \mapsto \mathcal{G}(\theta)$, as they in particular imply that the model $\{P_\theta : \theta \in \Theta\}$ of probability distributions is *identifiable*, i.e. $P_{\theta_1} = P_{\theta_2}$ can only occur whenever $\theta_1 = \theta_2$.

DEFINITION 1.3. The forward map $\mathcal{G}$ satisfies a *stability estimate* if, for some $\eta > 0$ and all $M > 0$, there exists a constant $L' = L'_{\mathcal{G}}$ such that, for all $\delta > 0$,

$$\sup\{\|\theta - \theta_0\|_{L^2} : \theta \in \Theta \cap B_{\mathcal{R}}(M), \|\mathcal{G}(\theta) - \mathcal{G}(\theta_0)\|_{L^2} \leq \delta\} \leq L' \delta^\eta.$$

Assuming such condition for Sobolev $\mathcal{R}$ and some $\eta, L' > 0$, we can choose m large enough such that:

$$P_{\theta_0}^N \left\{ \Pi_N(\theta \in \Theta_N, \|\theta - \theta_0\|_{L^2} \leq L'(m\delta_N)^\eta \mid D_N) \leq 1 - e^{-bN\delta_N^2} \right\} = o(1).$$

Putting all that precedes together, we can obtain posterior contraction theorems for the nonlinear inverse problem arising from some model PDE.

1.2. Posterior computation by MCMC. The nonlinearity of the forward map renders the posterior (II) analytically intractable: neither its normalising constant nor its moments admit closed-form expressions. One must therefore resort to MCMC techniques, the first historical instance of which was the Metropolis algorithm proposed in [29] to sample from the Boltzmann-Gibbs equilibrium measure

$$\begin{aligned} \rho_\beta(dq) &= \frac{1}{Z} e^{-\beta H(q)} dq \\ Z &= \int_{\mathbb{T}^d} e^{-\beta H(q)} dq \end{aligned}$$

of a system of n interacting particles with pairwise interaction energy given by Hamiltonians of the type

$$H(q^1, \dots, q^n) = \sum_{i=1}^n V(q^i) + \frac{1}{2n} \sum_{i=1}^n \sum_{j \neq i}^n W(q^i - q^j)$$

where V, W are some specified potentials and β the inverse temperature. In our case, when $\gamma = 0$, the dynamics underlying (IPS) become deterministic, driven by the mean-field Hamiltonian

$$H(\{q^i, p^i\}_{i=1}^n) = \frac{1}{2} \sum_{i=1}^n |p^i|^2 + \sum_{i=1}^n V(q^i) + \frac{1}{2n} \sum_{i=1}^n \sum_{j \neq i}^n W(q^i - q^j),$$

which is conserved along the flow; for $\gamma > 0$ and finite n , it is selected, as invariant distribution for the system, the Gibbs measure that has ρ_β as its spatial marginal.

The idea is to build a Markov process whose stationary measure is ρ_β : starting from the current configuration, a small random perturbation of a particle is proposed and accepted with probability $1 \wedge e^{-\beta \Delta H}$, where ΔH is the resulting change in energy. In particular, moves that ‘decrease the energy’ are always accepted, and the others are only admitted with acceptance probability $e^{-\beta \Delta H}$. At this point, for such a system at finite temperature, one can prove that the resulting Markov chain is irreducible and aperiodic, hence ergodic, and that the algorithm converges to the Boltzmann distribution; see [30]. Hastings extended this paradigm in [31] to more general

target distributions and non-symmetric proposal kernels, including systems without an energy. Even in such cases, when working with Gaussian priors, while it is likely that the posterior distribution is not itself Gaussian, it will at least be absolutely continuous with respect to some appropriate Gaussian density and, typically, the dominating Gaussian measure can be chosen to be exactly the prior, whence the Radon-Nikodym derivative is given by the Bayes formula:

$$\frac{d\Pi(\theta | \cdot)}{d\Pi(\theta)} = \frac{1}{Z} e^{\ell_N(\theta)}$$

$$Z = \int_{\Theta} e^{\ell_N(\theta)} d\Pi(\theta)$$

so that minus the log-likelihood can effectively be considered as a potential, and thus an energy functional, associated to the problem and the same framework naturally extends when one needs to construct Markov chains that are valued in some Banach space, as in our present context; see [32, 33, 34, 35, 36, 37].

In such an infinite-dimensional setting, however, a naive random walk Metropolis–Hastings scheme is not robust. A proposal of the form:

$$\vartheta_{k+1} = \vartheta_k + \sqrt{2\delta} \xi_k,$$

with $\xi_k \sim \Pi$ a draw from the centred Gaussian prior and fixed ‘step size’ $\delta > 0$, produces chains $(\vartheta_k)_k$ whose acceptance probabilities degenerate as the discretisation of Θ is refined, so that the algorithm slows down as the dimension of the numerical approximation grows. The preconditioned Crank–Nicolson (pCN) algorithm of [23] is a modification of Metropolis–Hastings designed to remove this deficiency:

$$\vartheta_{k+1} = \sqrt{1 - 2\delta} \vartheta_k + \sqrt{2\delta} \xi_k$$

and accepts the move with probability $1 \wedge e^{\ell_N(\vartheta_{k+1}) - \ell_N(\vartheta_k)}$. This corresponds to the autoregressive update already proposed in [17] for sampling latent values in classification models, and its shrinkage factor $\sqrt{1 - 2\delta}$ makes the proposal kernel reversible with respect to the Gaussian prior Π and it may be viewed as a semi-implicit Crank–Nicolson discretisation of a prior-preserving Ornstein–Uhlenbeck dynamics on Θ , so that, by the Bayes formula above, the acceptance ratio involves only the likelihood difference and the algorithm is well-defined directly on the function space. For the resulting nonparametric posteriors arising from PDE models such as (VFP), polynomial-time mixing bounds have been obtained in related settings by [24].

2. CONDITIONAL POSTERIOR CONTRACTION FOR THE KINETIC MODEL

In this section, we formulate the assumptions on the Vlasov–Fokker–Planck solution map under which the general posterior contraction theorem of Section 1 applies to the two measurement models i. and ii. described in the Introduction, and we state the resulting contraction theorems for the “forward” quantities ρ_W and f_W . The theorems are *conditional*: they take as hypotheses a well-posedness/regularity statement (Assumption 2.2) and an entropy stability estimate (Assumption 2.3) for the equation (VFP), which play the roles of Theorem 5 and Lemma 1 of [1] in the overdamped setting, and whose verification by hypoelliptic regularity and hypocoercivity techniques is deferred to future works.

For $k > 0$ and $s \geq 0$, we write $H_s^k(\mathbb{T}^d \times \mathbb{R}^d)$ for the weighted Sobolev space of functions u with

$$\|u\|_{H_s^k} \equiv \|\langle p \rangle^s u(q, p)\|_{H^k(\mathbb{T}^d \times \mathbb{R}^d)} < \infty;$$

cf. [20, Proof of Thm. 38]. Since the measurement noise in models i. and ii. has known variance σ_ϵ^2 , we may and do assume $\sigma_\epsilon = 1$ after replacing Y_i by Y_i/σ_ϵ and the forward maps by $\sigma_\epsilon^{-1} \rho_W, \sigma_\epsilon^{-1} f_W$; the constant σ_ϵ is absorbed into the envelope and Lipschitz constants below and does not affect the rates.

Recall that the interaction potential ranges over a class $\mathscr{W}$ of symmetric, mean-zero potentials with Lipschitz gradient. To apply Gaussian process techniques we require, as in (5) of [1], that $\mathscr{W}$ be realised as a separable

normed linear space satisfying the continuous embedding

$$\mathcal{W} \hookrightarrow C^2(\mathbb{T}^d) \cap \left\{ W : \int_{\mathbb{T}^d} W(q) dq = 0, W(-q) = W(q) \right\}. \quad (\text{W})$$

The mean-zero constraint encodes the identifiability normalisation introduced above, while the symmetry constraint reflects the physical symmetry $W(-q) = W(q)$ of the interaction and can be realised at the level of Fourier expansions by retaining only the cosine frequencies, cf. Remark 2 in [1].

CONDITION 2.1. Let $\tilde{\Pi}$ be a centred Gaussian Borel probability measure on a separable space $\mathcal{W}$ satisfying (W), with reproducing kernel Hilbert space $(\mathcal{H}, \|\cdot\|_{\mathcal{H}})$ satisfying, for some $\alpha > 1 + d/2$, the continuous embedding

$$\mathcal{H} \hookrightarrow H^{\alpha+1}(\mathbb{T}^d) \cap \left\{ W : \int_{\mathbb{T}^d} W(q) dq = 0, W(-q) = W(q) \right\}.$$

The prototypical example is the periodised Matérn process of Example 1 in [1] with the sine coefficients set to zero, whose RKHS is the symmetric mean-zero subspace of $H^{\alpha+1}(\mathbb{T}^d)$, and which for $\alpha > 1 + d/2$ defines a Gaussian random element of a separable Besov space $B'_{\infty,1}(\mathbb{T}^d) \hookrightarrow C^2(\mathbb{T}^d)$, $2 < r < \alpha + 1 - d/2$. The condition $\alpha > 1 + d/2$ guarantees in particular that draws lie in $C^2(\mathbb{T}^d) \hookrightarrow W^{2,\infty}(\mathbb{T}^d)$, the regularity needed for the PDE estimates assumed below; the space $\mathcal{R} = \mathcal{W} \hookrightarrow W^{2,\infty}(\mathbb{T}^d)$, with embedding constant $c > 0$, will serve as the regularisation space of Definition 1.1.

Given $\kappa > 0$ as in the assumptions below, we define the *rescaled prior*

$$\Pi_N = \text{Law} \left(\frac{\tilde{W}}{\sqrt{N}\delta_N} \right), \quad \delta_N = N^{-(\alpha+1+\kappa)/(2(\alpha+1)+2\kappa+d)} \quad \text{and} \quad \tilde{W} \sim \tilde{\Pi} \quad (\Pi_N)$$

noting that $\sqrt{N}\delta_N = N^{d/(4(\alpha+1)+4\kappa+2d)}$, so that (Π_N) coincides with the shrinkage chosen in Section 1. One may further compose with $L^2(\mathbb{T}^d)$ -projections $\pi_{\mathcal{W}_N}$ onto linear subspaces $\mathcal{W}_N \subseteq \mathcal{W}$ satisfying

$$\|\pi_{\mathcal{W}_N}(w)\|_{\mathcal{W}} \leq C\|w\|_{\mathcal{W}}$$

uniformly in N (e.g. truncation of the cosine expansion at frequency K_N), which will be relevant for the inverse problem; for the statements below we take $\mathcal{W}_N = \mathcal{W}$ and comment on the projected case in Remark 5.

2.1. Analytical assumptions on the forward maps. We now formulate the two structural hypotheses on (VFP). Throughout, we assume the initial condition

$$\phi \in L^\infty(\mathbb{T}^d \times \mathbb{R}^d) \cap H_s^{\kappa+1}(\mathbb{T}^d \times \mathbb{R}^d) \cap \mathcal{P}(\mathbb{T}^d \times \mathbb{R}^d)$$

for some $\kappa > 0$ and momentum weight $s > d/2$, the friction $\gamma > 0$, inverse temperature $\beta > 0$ and confining potential $V \in C^\infty(\mathbb{T}^d)$ are fixed and known, and all constants may depend on $(d, T, \gamma, \beta, V, \phi)$ without further mention.

ASSUMPTION 2.2 (WELL-POSEDNESS AND REGULARITY). For every $M > 0$ one can find constants $U = U(M) \geq 1$ and $C = C(M) > 0$ with the following properties, valid for every $W \in \mathcal{W}$ with $\|W\|_{W^{2,\infty}(\mathbb{T}^d)} \leq M$,

i) *Well-posedness*: equation (VFP) possesses a unique classical solution $f_W \geq 0$ on $\mathcal{X}$ with

$$f_W(t, \cdot, \cdot) \in \mathcal{P}(\mathbb{T}^d \times \mathbb{R}^d)$$

for every $t \in [0, T]$;

ii) *Phase-space envelope*: $\sup_{(t,q,p) \in \mathcal{X}} f_W(t, q, p) \leq U$;

iii) *Macroscopic envelope*: $\sup_{t \in [0, T]} \|\rho_W(t, \cdot)\|_{L^\infty(\mathbb{T}^d)} \leq U$;

iv) *Marginal Sobolev regularity*: $\|\rho_W\|_{L^2(0, T; H^{\kappa+1}(\mathbb{T}^d))} \leq C$.

ASSUMPTION 2.3 (ENTROPY STABILITY). For every $M > 0$ and all $W_1, W_2 \in \mathcal{W}$ with $\|W_i\|_{W^{2,\infty}(\mathbb{T}^d)} \leq M$, the corresponding solutions $f_i = f_{W_i}$ of (VFP) satisfy, for all $t \in [0, T]$,

$$H(f_1(t) | f_2(t)) \equiv \int_{\mathbb{T}^d \times \mathbb{R}^d} f_1(t) \log \frac{f_1(t)}{f_2(t)} \leq \frac{\beta}{4\gamma} \int_0^t \int_{\mathbb{T}^d \times \mathbb{R}^d} |[\nabla_q W_1 * \rho_1 - \nabla_q W_2 * \rho_2](s, q)|^2 f_1(s, q, p) \, dq \, dp \, ds, \quad (\text{H})$$

where $\rho_i = \rho_{W_i}$ denote the spatial marginals.

2.1.1. *On the plausibility of the assumptions.* Assumption 2.2 is the kinetic counterpart of Theorem 5 in [1]. Well-posedness i), uniformly over $W^{2,\infty}$ -balls of the interaction, is expected from the perturbative analysis of (VFP) in [10, 11]: the transport operator $p \cdot \nabla_q$ is not smoothing by itself, but the Hörmander-type interaction between transport and the momentum diffusion $\gamma \beta^{-1} \Delta_p$ produces hypoelliptic regularisation, and the hypocoercivity framework of [20, 21] provides the natural quantitative setting; when ϕ has Gaussian momentum decay, the Maxwellian-weighted spaces of [20] are the natural choice, matching the Gaussian design density ν discussed in the Introduction.

The phase-space envelope ii) would then hold essentially for free, with a constant independent of the interaction: expanding the friction term as $\gamma \nabla_p \cdot (pf) = \gamma df + \gamma p \cdot \nabla_p f$, at an interior maximum of $f_W(t, \cdot, \cdot)$ all first-order terms vanish, while $\Delta_p f \leq 0$, so that only the zeroth-order contribution γdf survives and the maximum principle yields

$$\sup_{(q,p)} f_W(t, q, p) \leq e^{\gamma dt} \|\phi\|_{L^\infty(\mathbb{T}^d \times \mathbb{R}^d)}, \quad t \in [0, T],$$

uniformly over all $W \in \mathcal{W}$, since the interaction enters only through the drift $\nabla_q(V + W * \rho) \cdot \nabla_p f$, which is of first order. The macroscopic envelope iii), by contrast, is not implied by ii): a uniform bound on f_W says nothing about the convergence of the momentum integral defining ρ_W . It is a genuine uniform momentum-decay statement, to be obtained from moment (Lyapunov) estimates propagating the decay encoded in $\phi \in H_s^{\kappa+1}$, $s > d/2$, along the flow.

Estimate iv) is a pure *propagation of regularity* statement, and this is where the strengthened hypothesis on ϕ intervenes. Note first that, for $k \in \mathbb{N}$, the norm $\|\cdot\|_{H_s^k}$ is equivalent to

$$\left(\sum_{|\sigma| \leq k} \|\langle p \rangle^s \partial^\sigma u(q, p)\|_{L^2(\mathbb{T}^d \times \mathbb{R}^d)}^2 \right)^{1/2}$$

where σ ranges over multi-indices in the variables (q, p) , since every derivative of $\langle p \rangle^s$ is bounded by a constant multiple of $\langle p \rangle^s$; for non-integer orders, $H_s^{\kappa+1}$ may equivalently be obtained by interpolation between the integer-order spaces. Writing

$$\hat{u}(k, p) = \int_{\mathbb{T}^d} u(q, p) e^{-2\pi i k \cdot q} \, dq$$

for the partial Fourier coefficients in the spatial variable, the Cauchy–Schwarz inequality in p gives, since $s > d/2$,

$$|\hat{\rho}(0, k)|^2 = \left| \int_{\mathbb{R}^d} \hat{\phi}(k, p) \, dp \right|^2 \leq \left(\int_{\mathbb{R}^d} \langle p \rangle^{-2s} \, dp \right) \int_{\mathbb{R}^d} \langle p \rangle^{2s} |\hat{\phi}(k, p)|^2 \, dp,$$

and multiplying by $\langle k \rangle^{2(\kappa+1)}$ and summing over $k \in \mathbb{Z}^d$ yields

$$\|\rho(0, \cdot)\|_{H^{\kappa+1}(\mathbb{T}^d)}^2 \lesssim_s \sum_{k \in \mathbb{Z}^d} \langle k \rangle^{2(\kappa+1)} \|\langle p \rangle^s \hat{\phi}(k, \cdot)\|_{L^2(\mathbb{R}^d)}^2 \leq \|\phi\|_{H_s^{\kappa+1}}^2,$$

where the last inequality holds because the full phase-space Sobolev norm dominates the mixed norm $L_p^2 H_q^{\kappa+1}$. Thus, the spatial marginal already satisfies $\rho(0, \cdot) \in H^{\kappa+1}(\mathbb{T}^d)$ at initial time, and iv) merely asserts that this regularity is propagated, in a time-averaged sense, along the flow of (VFP): no hypoelliptic gain of spatial derivatives is required. Such propagation is expected from Grönwall-type energy estimates in the spaces H_s^k at integer orders k , uniformly over $\{\|W\|_{W^{2,\infty}} \leq M\}$, followed by interpolation to reach fractional $\kappa + 1$, in the spirit of Theorem

38 in [20]. The mean-field term is harmless at any order of differentiation: since convolution in q commutes with every Fourier multiplier in q , one has $\langle D_q \rangle^{\kappa+1} (\nabla_q W * \rho) = \nabla_q W * \langle D_q \rangle^{\kappa+1} \rho$, so that only $\|W\|_{W^{2,\infty}}$ enters the constants regardless of the size of κ . Should the hypoelliptic transfer-of-regularity mechanisms additionally yield a fractional gain of spatial derivatives for the marginal, the hypothesis $\phi \in H_s^{\kappa+1}$ could be correspondingly weakened without affecting the statistical analysis, which depends on ϕ only through Assumption 2.2.

Assumption 2.3 is the kinetic analogue of Lemma 1 in [1], and its validity is strongly suggested by a Girsanov argument on (??) with the marginals ρ_i frozen: the drift perturbation $-\nabla_q[W_1 * \rho_1 - W_2 * \rho_2](t, \bar{q}_t)$ acts, like the noise, only on the momentum equation, as it lies in the range of the degenerate diffusion matrix, with Novikov's condition automatic since it is bounded by $2M$ on $\{\|W_i\|_{W^{2,\infty}} \leq M\}$; the path-space relative entropy then equals half the expected squared drift difference weighted by $(2\gamma\beta^{-1})^{-1}$, and (H) follows, with the constant $\beta/(4\gamma)$, by the data-processing inequality. No result in the literature states (H) itself, but each of its ingredients is established in an adjacent setting: [38] prove the two-solution estimate, with the same quadratic drift-difference functional, for uniformly nondegenerate Fokker–Planck–Kolmogorov equations, whence its direct use in the overdamped case; the weak-strong uniqueness theorem of [39] provides a two-solution entropy calculus for the degenerate kinetic operator, controlling the cross term by weighted Csiszár–Kullback–Pinsker inequalities rather than by absorption into the dissipation; the Girsanov method of [40] tolerates precisely this degeneracy but compares the particle system with its mean-field limit at a common interaction; and [41], a degenerate semigroup extension of [38], obtain (H) with the same constant and $\|\cdot\|_{L^\infty(\mathbb{T}^d)}$ by means of the f_1 -weighted norm, on $\mathbb{R}^{2d}$ under stronger coefficient regularity; cf. Appendix.

2.2. Forward regularity of the Vlasov–Fokker–Planck solution maps. Define the two forward maps

$$\mathcal{E}_\rho : W \mapsto \rho_W \in L^2(\mathcal{X}, \lambda), \quad \mathcal{E}_f : W \mapsto f_W \in L^2(\mathcal{X}, \lambda \otimes \nu).$$

The next lemma is the key analytical step: it verifies, conditionally on Assumptions 2.2–2.3, that both maps are forward regular in the sense of Definition 1.1 with smoothing index κ , the kinetic analogue of Lemma 3 in [1].

LEMMA 2.4 (CONDITIONAL FORWARD REGULARITY). Suppose Assumptions 2.2 and 2.3 are satisfied with index $\kappa > 0$, and let $M > 0$. Then there exists $L = L(M) > 0$ such that for all $W_1, W_2 \in \mathcal{W}$ with $\|W_i\|_{W^{2,\infty}} \leq M$,

$$\|\rho_1 - \rho_2\|_{L^2(\mathcal{X}, \lambda)} + \|f_{W_1} - f_{W_2}\|_{L^2(\mathcal{X}, \lambda \otimes \nu)} \leq L \|W_1 - W_2\|_{H^{-\kappa}(\mathbb{T}^d)}. \quad (\text{Reg})$$

In particular, in view of Assumption 2.2 ii)–iii), both $\mathcal{E}_\rho$ and $\mathcal{E}_f$ are forward regular with smoothing index κ and regularisation space $\mathcal{R} = \mathcal{W} \hookrightarrow W^{2,\infty}(\mathbb{T}^d)$.

Proof. Fix M and W_1, W_2 as in the statement, write $f_i = f_{W_i}$, $\rho_i = \rho_{W_i}$ and $H(t) = H(f_1(t)|f_2(t))$.

Step 1: Grönwall estimate for the relative entropy. Since the integrand in (H) depends on (s, q) only, integrating out p and using Assumption 2.2 iii),

$$H(t) \leq \frac{\beta}{4\gamma} \int_0^t \int_{\mathbb{T}^d} |\nabla_q W_1 * \rho_1 - \nabla_q W_2 * \rho_2|^2(s, q) \rho_1(s, q) dq ds \leq \frac{\beta U}{4\gamma} \int_0^t \|\nabla_q W_1 * \rho_1 - \nabla_q W_2 * \rho_2\|_{L^2(\mathbb{T}^d)}^2(s) ds.$$

Splitting $\nabla_q W_1 * \rho_1 - \nabla_q W_2 * \rho_2 = \nabla_q W_1 * (\rho_1 - \rho_2) + \nabla_q (W_1 - W_2) * \rho_2$ and using Young's convolution inequality,

$$\|\nabla_q W_1 * (\rho_1 - \rho_2)(s)\|_{L^2(\mathbb{T}^d)} \leq \|\nabla_q W_1\|_{L^2(\mathbb{T}^d)} \|\rho_1(s) - \rho_2(s)\|_{L^1(\mathbb{T}^d)} \leq M \|f_1(s) - f_2(s)\|_{L^1(\mathbb{T}^d \times \mathbb{R}^d)} \leq M \sqrt{2H(s)},$$

where we used that the total variation distance is non-increasing under marginalisation, followed by Pinsker's inequality [28, Proposition 6.1.7]. Combining the two previous inequalities gives, for $0 \leq t \leq T$,

$$\dot{\eta}(t) \leq \frac{\beta U}{2\gamma} \left(2M^2 \eta(t) + \|\nabla_q (W_1 - W_2) * \rho_2\|_{L^2(\mathcal{X})}^2 \right) \quad \text{where } \eta(t) \equiv \int_0^t H(s) ds$$

and the differential form of Grönwall's inequality yields, for some constant $G = G(M) > 0$,

$$\int_0^T H(t) dt \leq G \|\nabla_q(W_1 - W_2) * \rho_2\|_{L^2(\mathcal{X})}^2. \quad (\text{G})$$

Step 2: deconvolution. Exactly as in the proof of Lemma 3 (ii) in [1], Parseval's theorem gives, pointwise in t ,

$$\|\nabla_q(W_1 - W_2) * \rho_2(t, \cdot)\|_{L^2(\mathbb{T}^d)}^2 \leq 4\pi^2 \|\rho_2(t, \cdot)\|_{H^{1+\kappa}(\mathbb{T}^d)}^2 \|W_1 - W_2\|_{H^{-\kappa}(\mathbb{T}^d)}^2,$$

whence, integrating in time and invoking Assumption 2.2 iv),

$$\|\nabla_q(W_1 - W_2) * \rho_2\|_{L^2(\mathcal{X})}^2 \leq 4\pi^2 C(M)^2 \|W_1 - W_2\|_{H^{-\kappa}(\mathbb{T}^d)}^2. \quad (\text{P})$$

Step 3: from entropy to the design norms. Let

$$h^2(u_1, u_2) = \int_{\mathbb{T}^d \times \mathbb{R}^d} (\sqrt{u_1} - \sqrt{u_2})^2$$

denote the squared *Hellinger distance* on $\mathbb{T}^d \times \mathbb{R}^d$ and recall, e.g. from [27, Lemma B.1], that $h^2 \leq H$. For the phase-space model, writing $(f_1 - f_2)^2 = (\sqrt{f_1} + \sqrt{f_2})^2 (\sqrt{f_1} - \sqrt{f_2})^2$ and using $\|v\|_\infty < \infty$ together with Assumption 2.2 ii),

$$\|f_1(t) - f_2(t)\|_{L^2(\mathbb{T}^d \times \mathbb{R}^d, dq \otimes v dp)}^2 \leq 4 \|v\|_\infty U h^2(f_1(t), f_2(t)) \leq 4 \|v\|_\infty U H(t).$$

For the macroscopic model, by the Cauchy–Schwarz inequality in the momentum variable,

$$(\rho_1 - \rho_2)^2 = \left(\int_{\mathbb{R}^d} (\sqrt{f_1} + \sqrt{f_2}) (\sqrt{f_1} - \sqrt{f_2}) (\cdot, \cdot, p) dp \right)^2 \leq 2(\rho_1 + \rho_2) \int_{\mathbb{R}^d} (\sqrt{f_1} - \sqrt{f_2})^2 (\cdot, \cdot, p) dp,$$

so that, integrating in q and using Assumption 2.2 iii),

$$\|\rho_1(t) - \rho_2(t)\|_{L^2(\mathbb{T}^d)}^2 \leq 4U h^2(f_1(t), f_2(t)) \leq 4U H(t).$$

Averaging over $t \sim \lambda_T$, and combining with (G)–(P), proves (Reg). The uniform boundedness required by Definition 1.1 i) is precisely Assumption 2.2 ii) for $\mathcal{G}_f$ and iii) for $\mathcal{G}_\rho$, both uniform over $\{\|W\|_{W^{2,\infty}} \leq M\} \supseteq \mathcal{W} \cap B_{\mathcal{R}}(M/c)$ by (W). $\blacksquare$

REMARK 2. The conditional forward regularity Lemma 2.4 shows that the statistical ill-posedness of both measurement models is governed by κ , the Sobolev regularity of the initial condition, exactly as in the overdamped case: the map $W \mapsto \rho_W$ (resp. $W \mapsto f_W$) is κ -smoothing because W enters (VFP) only through the spatial convolution $\nabla_q W * \rho$, which trades regularity of W against that of ρ_W . The kinetic structure affects only the constants (through β/γ , the envelopes U , and $\|v\|_\infty$), not the smoothing index.

2.3. Posterior contraction theorems. We can now state the two conditional contraction theorems, the analogues of Theorem 1 in [1]. Recall the posterior (II) built from the rescaled prior (II_N) and the data $D_N = (Y_i, t_i, q_i)_{i=1}^N$ (model i.) or $D_N = (Y_i, t_i, q_i, p_i)_{i=1}^N$ (model ii.), and write $P_{W_0}^N$ for the law of D_N under the ground truth W_0 .

Given $M > 0$ and κ as above, define the regularisation sets

$$\mathcal{W}_{M,N} = \left\{ W = W_1 + W_2 \in \mathcal{W} : \|W_1\|_{H^{-\kappa}(\mathbb{T}^d)} \leq M\delta_N, \|W_2\|_{\mathcal{H}} \leq M, \|W\|_{C^2(\mathbb{T}^d)} \leq M \right\}.$$

THEOREM 2.5 (MACROSCOPIC OBSERVATIONS). Let $\bar{\Pi}$ satisfy Condition 2.1 with $\alpha > 1 + d/2$. Under Assumptions 2.2 and 2.3, suppose the data D_N arise from measurement model i. with ground truth $W_0 \in \mathcal{H}$. Then, for $M, L > 0$ large enough,

$$\Pi_N \left(W \in \mathcal{W}_{M,N} : \|\rho_W - \rho_{W_0}\|_{L^2(\mathcal{X}, \lambda)} \leq L\delta_N \mid D_N \right) \xrightarrow{P_{W_0}^N} 1,$$

as $N \rightarrow \infty$.

THEOREM 2.6 (PHASE-SPACE OBSERVATIONS). Under the hypotheses of Theorem 2.5, suppose instead that D_N arises from measurement model ii. with ν a bounded, strictly positive probability density on $\mathbb{R}^d$. Then, for $M, L > 0$ large enough,

$$\Pi_N \left(W \in \mathcal{W}_{M,N} : \|f_W - f_{W_0}\|_{L^2(\mathcal{X}, \lambda \otimes \nu)} \leq L\delta_N \mid D_N \right) \xrightarrow{P_{W_0}^N} 1,$$

as $N \rightarrow \infty$.

Proof of Theorems 2.5 and 2.6. We verify the hypotheses of Theorem 1.2, in the generality of Remark 1, to the triplet $(\Theta, \mathcal{R}, \mathcal{G})$ with $\Theta = \mathcal{W}$, $\mathcal{R} = \mathcal{W}$, and $\mathcal{G}$ equal to $\mathcal{G}_\rho$ (model i.) or $\mathcal{G}_f$ (model ii.), the regularity parameter being $\alpha + 1$ and the smoothing index κ , so that the rescaling in (Π_N) coincides with the shrinkage prescribed there and $\delta_N = N^{-(\alpha+1+\kappa)/(2(\alpha+1)+2\kappa+d)}$ is the associated contraction rate. Throughout, $c \geq 1$ denotes the embedding constant of (W), so that $\|W\|_{C^2(\mathbb{T}^d)} \leq c\|W\|_{\mathcal{W}}$ for all $W \in \mathcal{W}$.

Step 1: the ground truth. Since $\bar{\Pi}$ is a centred Gaussian Borel probability measure on the separable space $\mathcal{W}$, its RKHS is continuously embedded into (the closure of the support of $\bar{\Pi}$ in) $\mathcal{W}$; see [28, Theorem 2.6.9]. Hence $\mathcal{H} \subseteq \mathcal{W} = \mathcal{R}$ and the hypothesis $W_0 \in \mathcal{H}$ of the theorems implies $W_0 \in \mathcal{H} \cap \mathcal{R}$, as required by Theorem 1.2. By Condition 2.1, $\bar{\Pi}(\mathcal{R}) = 1$ and $\mathcal{H} \hookrightarrow H^{\alpha+1}(\mathbb{T}^d)$ intersected with the mean-zero symmetric constraint, which is the embedding hypothesis of Theorem 1.2 in the form of Remark 1.

Step 2: forward regularity. Let $M > 0$. Every $W \in \mathcal{W} \cap B_{\mathcal{R}}(M)$ satisfies $\|W\|_{W^{2,\infty}(\mathbb{T}^d)} \leq \|W\|_{C^2(\mathbb{T}^d)} \leq cM$, so Lemma 2.4, applied with radius cM , yields the Lipschitz estimate of Definition 1.1 ii) with constant $L(cM)$, in the norm $H^{-\kappa}(\mathbb{T}^d) = (H^\kappa(\mathbb{T}^d))^*$; the uniform bound of Definition 1.1 i) holds with envelope $U(cM)$ by Assumption 2.2 iii) for $\mathcal{G}_\rho$ and ii) for $\mathcal{G}_f$, the latter measured in $L^\infty(\mathcal{X})$ and hence in particular $(\lambda \otimes \nu)$ -essentially bounded. Both maps are therefore forward regular with smoothing index κ , regularisation space $\mathcal{R} = \mathcal{W}$, and constants $U(c \cdot), L(c \cdot)$. Moreover, by (Reg), $\mathcal{G}$ is continuous from $(\mathcal{W}, \|\cdot\|_{H^{-\kappa}})$, hence Borel measurable on $\mathcal{W}$, into the respective design L^2 space. As for joint measurability of $(W, x) \mapsto \mathcal{G}(W)[x]$: the classical solution f_W of Assumption 2.2 i) is continuous in the design variable, and ρ_W , obtained from f_W by integration in p , is jointly measurable in (t, q) by Fubini-Tonelli; combined with the $H^{-\kappa}$ -continuity in W just noted, the maps $(W, x) \mapsto \mathcal{G}(W)[x]$ and $W \mapsto \ell_N(W)$ are jointly, respectively Borel, measurable, so that the posterior (Π) is well defined; cf. [2, Sec. 1.2.3].

Step 3: conclusions of Theorem 1.2. Set $M_0 \equiv \|W_0\|_{\mathcal{R}} \vee 1$ and let $U \equiv U(cM_0)$ be the associated envelope. Theorem 1.2 furnishes constants A, B with $B > A + 2$, sets

$$\Theta_N = \left\{ W \in \mathcal{R} : W = W_1 + W_2, \|W_1\|_{H^{-\kappa}(\mathbb{T}^d)} \leq M\delta_N, \|W_2\|_{\mathcal{H}} \leq M, \|W\|_{\mathcal{R}} \leq M \right\}$$

for some $M = M(B) > 0$, satisfying the small-ball, complement and metric entropy bounds stated there, and, for every $0 < b < B - A - 2$, a constant $L_0 = L_0(A, B, \bar{m}, b, U) > 0$ (absorbing the factor C_U^{-1}) such that

$$P_{W_0}^N \left\{ \Pi_N (W \in \Theta_N, d_{\mathcal{G}}(W, W_0) \leq L_0\delta_N \mid D_N) \leq 1 - e^{-bN\delta_N^2} \right\} \rightarrow 0, \quad N \rightarrow \infty. \quad (\text{PCT})$$

Step 4: from Θ_N to $\mathcal{W}_{\bar{M},N}$. Set $\bar{M} \equiv (1 \vee c)M$. If $W \in \Theta_N$, then W admits the decomposition $W = W_1 + W_2$ with $\|W_1\|_{H^{-\kappa}} \leq M\delta_N \leq \bar{M}\delta_N$ and $\|W_2\|_{\mathcal{H}} \leq M \leq \bar{M}$, and moreover $\|W\|_{C^2(\mathbb{T}^d)} \leq c\|W\|_{\mathcal{R}} \leq cM \leq \bar{M}$. Hence

$$\Theta_N \subseteq \mathcal{W}_{\bar{M},N},$$

and therefore, for every realisation of the data,

$$\Pi_N (W \in \mathcal{W}_{\bar{M},N}, d_{\mathcal{G}}(W, W_0) \leq L_0\delta_N \mid D_N) \geq \Pi_N (W \in \Theta_N, d_{\mathcal{G}}(W, W_0) \leq L_0\delta_N \mid D_N)$$

by monotonicity of the posterior measure.

Step 5: conclusion. Combining Step 4 with (PCT), and noting that $e^{-bN\delta_N^2} \rightarrow 0$ since $N\delta_N^2 \rightarrow \infty$, we obtain

$$\Pi_N (W \in \mathcal{W}_{\bar{M},N} : d_{\mathcal{G}}(W, W_0) \leq L_0 \delta_N \mid D_N) \xrightarrow{P_{W_0}^N} 1,$$

which is the claim with the constants (M, L) of the statements taken as $(\bar{M}, L_0)$, since $d_{\mathcal{G}}(W, W_0) = \|\rho_W - \rho_{W_0}\|_{L^2(\mathcal{X}, \lambda)}$ for $\mathcal{G} = \mathcal{G}_\rho$ and $d_{\mathcal{G}}(W, W_0) = \|f_W - f_{W_0}\|_{L^2(\mathcal{X}, \lambda \otimes \nu)}$ for $\mathcal{G} = \mathcal{G}_f$. $\blacksquare$

The envelope constants $U_N \equiv U(M)$ may be taken independent of N here.

REMARK 3 (ON THE DIMENSION AND THE KINETIC STRUCTURE). Although the phase-space state f_W in model ii. is a function of $2d$ spatial-type variables, the contraction rate δ_N features the torus dimension d only. This is because the rate is driven by the complexity of the parameter W , a function on $\mathbb{T}^d$, and by the smoothing index κ of the forward map, while the momentum variable enters the analysis only through multiplicative constants. In this sense, at the level of forward contraction rates, observing the full kinetic state is statistically no harder than observing its macroscopic marginal; the two models differ in the norm in which recovery is guaranteed.

REMARK 4 (ON THE CHOICE OF THE MOMENTUM DESIGN ν). The constant L in (Reg), hence the constants in Theorem 2.6, depend on ν only through $\|\nu\|_\infty$, so any bounded strictly positive design density yields the same rate δ_N . The choice of ν instead calibrates the meaning of the recovery guarantee: a heavy-tailed choice $\nu(p) \propto \langle p \rangle^{-2s}$, $s > d/2$, spreads statistical weight into the momentum tails where f_W is exponentially small, whereas a Gaussian ν concentrates the guarantee on the informative low-momentum region and aligns the design norm with the Maxwellian-weighted spaces of [20, 21] in which Assumption 2.2 is most naturally verified.

REMARK 5 (ON PROJECTED PRIORS). As in Example 2 of [1], Theorems 2.5–2.6 continue to hold, with the same rate, for the prior (Π_N) projected onto the span E_{K_N} of the symmetric trigonometric polynomials (cosine frequencies) $0 < |k| \leq K_N$ with $K_N \gtrsim N^{1/(2(\alpha+1)+2\kappa+d)}$, provided the truth admits an approximation sequence $W_{0,N} \in \mathcal{H} \cap E_{K_N}$ with $\|W_{0,N}\|_{\mathcal{H}} = O(1)$ and $\|W_0 - W_{0,N}\|_{H^{-\kappa}} = O(\delta_N)$, automatic for $W_0 \in \mathcal{H}$ by taking Fourier truncations. Such band-limited priors will be essential for the inverse problem of recovering W_0 itself, where, as anticipated in Section 1, a deconvolvability condition on ϕ — i.e. a quantitative lower bound on its Fourier coefficients in the spatial frequencies, ruling out initialisation at a stationary (in particular, spatially homogeneous) state of (VFP) — will be required to convert the forward contraction of Theorems 2.5–2.6 into contraction for $\|W - W_0\|_{L^2(\mathbb{T}^d)}$ via a stability estimate in the spirit of Theorem 2 of [1].

3. NUMERICAL EXPERIMENTS

In this section we implement the Bayesian inference scheme of Section 2 for both measurement models and gather numerical evidence for the conditional hypotheses underlying Theorems 2.5 and 2.6. The experiments serve the following purposes: they exhibit a structure-preserving forward solver for (VFP) that is accurate and fast enough to be called thousands of times inside a function-space MCMC loop, and whose spectral formulation makes the hypocoercive mechanism discussed after Assumption 2.2 directly visible as a matrix eigenvalue computation; they test empirically the two analytical inputs of the theory: the local Lipschitz continuity of the solution maps asserted, conditionally, by Lemma 2.4, and the envelope and regularity bounds of Assumption 2.2; they probe the structure of the problem beyond what the forward contraction theorems assert: by linearising the forward maps at the ground truth we quantify, mode by mode, which Fourier frequencies of W the data identify, we locate the crossover at which the posterior stops learning and reverts to the prior, and we compare the two measurement models head to head, providing quantitative support for the heuristic of Remark 3 that observing the full kinetic state carries essentially no more information about W than observing its macroscopic marginal.

Throughout, we work in dimension $d = 1$, so that phase space is the cylinder $(q, p) \in \mathbb{T} \times \mathbb{R}$ with $\mathbb{T} = [0, 2\pi)$, and we fix the friction and inverse temperature $\gamma = \beta = 1$, the known confining potential $V(q) = -\cos q$, and the time horizon $T = 2$.

3.1. A Hermite–Fourier spectral discretisation of (VFP). The dominant computational cost of the Bayesian method is the repeated evaluation of the log-likelihood $\ell_N(W)$, each instance of which requires one forward solve of (VFP). The discretisation must therefore be economical, and it should respect the structural features of the equation, i.e. conservation of mass, the Gaussian momentum equilibrium, and the degenerate, momentum-only dissipation; both requirements are satisfied by a spectral method adapted to the kinetic Fokker–Planck operator, in the tradition of velocity-Hermite (moment) expansions going back to [42]; see [43] for Vlasov-type transport and [44] for a survey of spectral and splitting schemes for kinetic equations.

3.1.1. The Maxwellian-adapted Hermite basis. Let $M(p) = \sqrt{\beta/2\pi} e^{-\beta p^2/2}$ denote the Maxwellian and define, for $n \in \mathbb{N}$,

$$e_n(p) = \frac{\sqrt{\beta}}{\sqrt{2\pi} n!} \text{He}_n(\sqrt{\beta} p) e^{-\beta p^2/2}, \quad (e_n)$$

where He_n are the probabilists' Hermite polynomials: $\text{He}_0 = 1$, $\text{He}_1(\xi) = \xi$, $\text{He}_{n+1}(\xi) = \xi \text{He}_n(\xi) - n \text{He}_{n-1}(\xi)$. Using the classical orthogonality

$$\int_{\mathbb{R}} \text{He}_n \text{He}_m e^{-\xi^2/2} d\xi = \sqrt{2\pi} n! \delta_{nm},$$

a change of variables $\xi = \sqrt{\beta} p$ shows that the family $(e_n)_{n \geq 0}$ is orthonormal in the *Maxwellian-weighted* space $L^2(M^{-1}(p)dp)$,

$$\int_{\mathbb{R}} e_n(p) e_m(p) \frac{dp}{M(p)} = \delta_{nm},$$

which is precisely the flat-space counterpart of the weighted L^2 spaces in which the hypocoercivity analysis of (VFP) is naturally set [20, 21] and which motivated the Gaussian choice of the momentum design density ν in the **Introduction**. Expanding the solution in this basis,

$$f(t, q, p) = \sum_{n=0}^{N_H-1} c_n(t, q) e_n(p), \quad c_n(t, q) = \int_{\mathbb{R}} f(t, q, p) e_n(p) \frac{dp}{M(p)}, \quad (\text{HF})$$

the first two coefficients are exactly the macroscopic observables of the problem:

$$\rho(t, q) = \int_{\mathbb{R}} f(t, q, p) dp = c_0(t, q), \quad j(t, q) \equiv \int_{\mathbb{R}} p f(t, q, p) dp = \frac{c_1(t, q)}{\sqrt{\beta}},$$

the momentum flux identity following from the ladder relation below. In particular, the density of measurement model i. is read off with no numerical momentum integration whatsoever, and c_1 is, up to the factor $\beta^{-1/2}$, the local mean momentum, the first quantity through which the force, and hence W , acts on the density.

This basis diagonalises the Ornstein–Uhlenbeck (OU) operator:

$$\gamma \partial_p (p e_n + \beta^{-1} \partial_p e_n) = -\gamma n e_n, \quad (\text{OU})$$

i.e. the $(e_n)_n$ are the eigenfunctions of the Fokker–Planck form of the generator with the exact spectrum $\{-\gamma n\}_{n \geq 0}$ [45, Ch. 4]; and multiplication by p and differentiation in p act as ladder (shift) operators,

$$p e_n = \frac{1}{\sqrt{\beta}} (\sqrt{n+1} e_{n+1} + \sqrt{n} e_{n-1}), \quad \partial_p e_n = -\sqrt{\beta(n+1)} e_{n+1}. \quad (\text{Lad})$$

3.1.2. The coefficient hierarchy. Inserting (HF) into (VFP), written as $\partial_t f = -p \partial_q f - F \partial_p f + \gamma \partial_p (p f + \beta^{-1} \partial_p f)$ with the *mean-field force*

$$F(t, q) = -\partial_q (V + W * \rho)(t, q), \quad \rho = c_0,$$

by (Lad), the transport term contributes

$$-p\partial_q f = -\sum_{n \geq 0} \partial_q c_n \frac{1}{\sqrt{\beta}} (\sqrt{n+1} e_{n+1} + \sqrt{n} e_{n-1}),$$

the force term contributes

$$-F\partial_p f = F \sum_{n \geq 0} c_n \sqrt{\beta(n+1)} e_{n+1},$$

and the friction term is diagonal by (OU). Collecting the coefficient of e_n yields the closed hierarchy of one-dimensional equations

$$\partial_t c_n = \underbrace{-\frac{1}{\sqrt{\beta}} (\sqrt{n} \partial_q c_{n-1} + \sqrt{n+1} \partial_q c_{n+1})}_{\text{streaming}} + \underbrace{\sqrt{\beta n} F(t, q) c_{n-1}}_{\text{mean-field force}} - \underbrace{\gamma n c_n}_{\text{friction}}, \quad 0 \leq n < N_H, \quad (\text{S})$$

with the convention $c_{-1} \equiv 0$ and the closure $c_{N_H} \equiv 0$ at the top level.

We now emphasise three structural remarks, each of which will matter for the statistics.

(a) *The continuity equation is the $n = 0$ row.* Since $\sqrt{\beta \cdot 0} = 0$ and $\gamma \cdot 0 = 0$, the density equation reduces to $\partial_t c_0 = -\beta^{-1/2} \partial_q c_1$, i.e. $\partial_t \rho + \partial_q j = 0$: the force and the friction never act on the density directly. All the information about W enters at levels $n \geq 1$ and reaches the observable ρ only through the streaming coupling $c_1 \rightarrow c_0$: an observation to which we return in Sections 3.1.5 and 3.7.

(b) *The nonlinearity is confined to one scalar field.* The unknown W , the sole infinite-dimensional parameter, influences (S) exclusively through the single function $F(t, q)$, which acts by moving mass up one Hermite level. In Fourier space, the convolution is diagonal: $\mathcal{F}[W * \rho](k) = \mathcal{F}[W](k) \mathcal{F}[\rho](k)$, so a candidate W enters the solver only through its cosine coefficients: this matches the parametrisation of the prior in Section 3.4.

(c) *Truncation is benign in the presence of friction.* The closure $c_{N_H} = 0$ is a spectral truncation; since ϕ below has Gaussian momentum decay, its Hermite coefficients decay super-algebraically and the friction term damps level n at rate γn , so energy reaching the top of the hierarchy is dissipated rather than reflected. This suppresses the “recurrence” and “filamentation” phenomena known for collisionless ($\gamma = 0$) Hermite–Vlasov schemes [43]; the checks of Section 3.1.4 confirm that $N_H = 14$ –20 modes suffice in the present regime.

3.1.3. Time discretisation: Strang splitting with exact substeps. We discretise q on the uniform grid of N_q points and pass to discrete Fourier coefficients $\hat{c}_n(t, k)$. The right-hand side of (S) splits as $L + \mathcal{N}(t)$ into its W -independent linear part (streaming and friction), which in Fourier space decouples across wavenumbers into the tridiagonal-plus-diagonal generators

$$(L_k c)_n = -\gamma n c_n - \frac{ik}{\sqrt{\beta}} (\sqrt{n} c_{n-1} + \sqrt{n+1} c_{n+1}), \quad k \in \mathbb{Z}, |k| \leq N_q/2, \quad (\text{L})$$

and the force part, a multiplication operator in q acting as the strictly lower-triangular (in n) generator

$$(\mathcal{N}c)_n = \sqrt{\beta n} F(q) c_{n-1}.$$

One time step of length Δt is the Strang composition [46]

$$c(t + \Delta t) = e^{\mathcal{N}\Delta t/2} e^{\Delta t L} e^{\mathcal{N}\Delta t/2} c(t) + O(\Delta t^3),$$

which is globally second-order accurate; the two occurrences of $\mathcal{N}$ are evaluated with the force F recomputed from the current density c_0 , so that the self-consistent nonlinearity is refreshed twice per step.

Finally, the scheme conserves mass exactly, not merely to consistency order: total mass is the $k = 0$ Fourier coefficient of c_0 ; the force kick does not touch the level $n = 0$ (the factor $\sqrt{\beta n}$ vanishes), and $e^{\Delta t L_0} = \text{diag}(1, e^{-\gamma \Delta t}, e^{-2\gamma \Delta t}, \dots)$ fixes the $(k, n) = (0, 0)$ coefficient. The measured mass drift over full nonlinear runs is $\leq 5 \times 10^{-15}$, i.e. machine rounding.

3.1.4. *Validation of the solver.* Four independent tests, each targeting a different component of the scheme, establish its correctness before any statistics are attempted.

(1) *Exact friction spectrum.* For spatially uniform data the streaming term is inactive, the dynamics reduce to the $k = 0$ block, and each Hermite coefficient must decay at the exact Ornstein–Uhlenbeck rate, $c_n(t) = e^{-\gamma n t} c_n(0)$, by (OU). Initialising $c_1, c_2, c_3 \neq 0$ and comparing against the exact exponentials gives maximal relative errors 3.8×10^{-16} , 2.4×10^{-15} and 7.1×10^{-16} for $n = 1, 2, 3$: the momentum dissipation is represented without discretisation error.

(2) *Exact mass conservation*, as discussed above: drift $\leq 5 \times 10^{-15}$ along nontrivial nonlinear trajectories.

(3) *Stationarity of the kinetic equilibrium and second-order consistency.* The stationary states of (VFP) are the local Maxwellians $f_\infty(q, p) = \rho_\infty(q)M(p)$ whose density solves the self-consistent Gibbs relation

$$\rho_\infty = \frac{e^{-\beta(V+W*\rho_\infty)}}{\int_{\mathbb{T}} e^{-\beta(V+W*\rho_\infty)}},$$

the spatial marginal of the mean-field Gibbs measure discussed in Section 1.2; we compute ρ_∞ by Picard iteration of this fixed-point equation to tolerance 10^{-14} . Fed the exact equilibrium, the solver must sit still; the residual drift $\max_{t \leq 1.6} \|c(t) - c_{\text{eq}}\|_\infty$ measures the pure splitting error. Halving the step repeatedly gives

$$\Delta t = 0.08 : 4.77 \times 10^{-4}, \quad 0.04 : 1.19 \times 10^{-4}, \quad 0.02 : 2.99 \times 10^{-5}, \quad 0.01 : 7.46 \times 10^{-6},$$

each refinement dividing the error by 4.00. This is the $O(\Delta t^2)$ signature of Strang splitting, confirming the twice-per-step refreshing of the self-consistent force introduces no first-order error.

(4) *Independent particle cross-check.* The most stringent test simulates the underlying interacting particle system (IPS) directly, sharing no numerical machinery with the spectral solver. We evolve $n = 2 \times 10^5$ particles by the semi-implicit (symplectic) Euler–Maruyama scheme

$$p_{m+1}^i = p_m^i + \Delta t \left[-\partial_q V(q_m^i) - \left(\partial_q W * \rho_m^{\text{hist}} \right) (q_m^i) - \gamma p_m^i \right] + \sqrt{2\gamma\beta^{-1}\Delta t} \xi_m^i, \quad q_{m+1}^i = q_m^i + \Delta t p_{m+1}^i,$$

with $\xi_m^i \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$, where the mean-field convolution is evaluated against the empirical spatial histogram ρ_m^{hist} of the current particle positions, i.e. the genuine n -body dynamics with the empirical measure in place of the law. Initial conditions are sampled i.i.d. from $\phi = \rho_0 \otimes M$ (positions by inversion of the c.d.f. of ρ_0 , momenta as Gaussians), matching the chaoticity assumption of the Introduction. At $t = T$ the particle histogram and the spectral density agree to $\|\rho^{\text{hist}} - \rho^{\text{PDE}}\|_{L^2(\mathbb{T})} = 0.019$ against the density scale $(2\pi)^{-1} \approx 0.16$, a few per cent, consistent with the Monte Carlo noise of a 48-bin histogram of 2×10^5 samples, while both have moved far from the initial profile; see Figure 1. Since the two computations approximate the same object through entirely different mathematics (a mean-field PDE versus an interacting SDE system), their agreement validates the solver and modelling simultaneously.

3.1.5. *Hypocoercivity in the discretised generator.* In each of the blocks (L), friction contributes the real, diagonal part $-\gamma \text{diag}(0, 1, 2, \dots)$, which damps every Hermite level except the density level $n = 0$; streaming contributes the purely imaginary, tridiagonal part $-ik\sqrt{\beta}^{-1}(\sqrt{n}c_{n-1} + \sqrt{n+1}c_{n+1})$, which is skew-adjoint and dissipates nothing. The long-time behaviour of the linear semigroup at spatial frequency k is governed by the spectral abscissa of L_k , and the numerically computed spectra (Figure 2) exhibit the dichotomy that the hypocoercivity theory of [20, 21] predicts.

At $k = 0$, the streaming ‘switches off’ and $L_0 = -\gamma \text{diag}(0, 1, 2, \dots)$ is the exact OU spectrum: the eigenvalue 0, whose eigenvector is the pure density mode, expresses mass conservation and shows that no coercive (uniform-in- k , including $k = 0$) decay estimate can hold. For $k \neq 0$, however, the streaming coupling of c_0 to the friction-damped level c_1 drags the whole spectrum strictly into the open left half-plane. Quantitatively, with $N_H = 14$

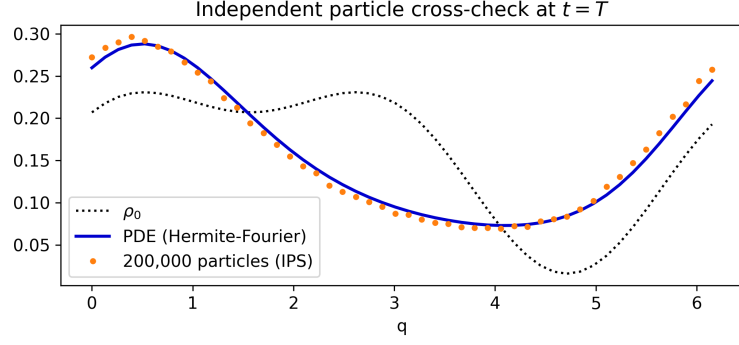


FIGURE 1. Independent particle cross-check. The spatial density at $t = T$ computed by the Hermite-Fourier solver (solid line) against the histogram of 2×10^5 particles of the interacting system (IPS) (dots), both started from the same initial law $\phi = \rho_0 \otimes M$ (dotted line). The two computations share no code and agree to within the Monte Carlo histogram noise, numerically instantiating the mean-field limit of the [Introduction](#).

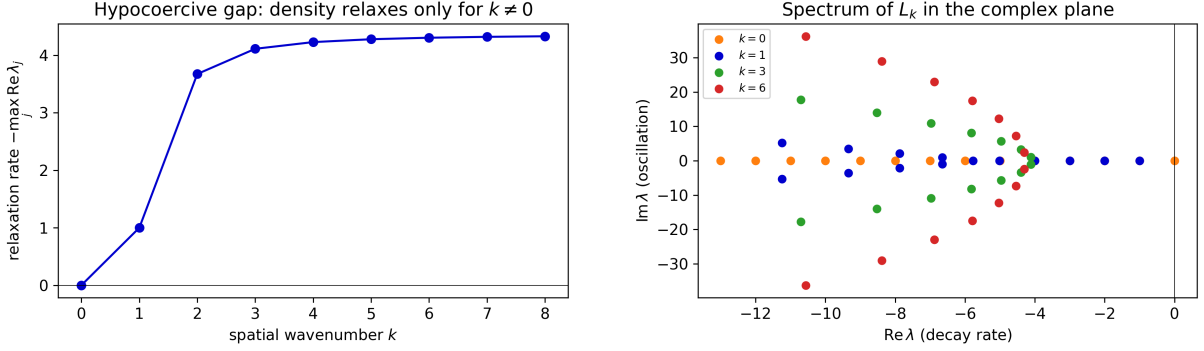


FIGURE 2. Left: least-damped relaxation rate $-\max_j \operatorname{Re} \lambda_j(L_k)$ against the spatial wavenumber k . At $k = 0$ the density mode is conserved (rate 0); at $|k| = 1$ a gap equal to γ opens, purely through the streaming coupling of the density to the friction-damped momentum modes, and saturates near 4.3γ . Right: the full spectrum of L_k in the complex plane for $k \in \{0, 1, 3, 6\}$. At $k = 0$ (red) the eigenvalues are the exact Ornstein-Uhlenbeck values $0, -\gamma, \dots, -13\gamma$ on the real axis; for $k \neq 0$ every eigenvalue moves strictly left of the imaginary axis while acquiring an imaginary (oscillatory) part.

and $\gamma = \beta = 1$, the computed relaxation rate $-\max_j \operatorname{Re} \lambda_j(L_k)$ equals

$$k = 0 : 0, \quad |k| = 1 : 1.000, \quad |k| = 2 : 3.674, \quad |k| = 3 : 4.112, \quad |k| \geq 4 : \nearrow \approx 4.33,$$

so that a spectral gap of size γ (to numerical precision) opens already at the first nonzero frequency and saturates at a few multiples of γ thereafter. In the complex plane (Figure 2, right) one watches the transition: at $k = 0$ the eigenvalues sit on the real axis at $0, -\gamma, -2\gamma, \dots$; as $|k|$ grows, they acquire imaginary parts of size $O(k/\sqrt{\beta})$, while their real parts detach uniformly from the imaginary axis.

We stress two consequences for the statistical problem that will be corroborated quantitatively below. First, this streaming-mediated relaxation is the mechanism that a proof of Assumption 2.2 would have to quantify: the overdamped model of [1] enjoys direct parabolic damping of the density at every wavenumber, whereas here the density is controlled only through its coupling to the momentum hierarchy, and estimates must be run in twisted norms of the type constructed in [20, 21]. Second, the same off-diagonal coupling is the *channel of identifiability*: by remark (a) of Section 3.1, the interaction acts only at levels $n \geq 1$, so it is precisely the streaming feedback $c_1 \rightarrow c_0$ that writes the imprint of W into the observable density. The identifiability analysis of Section 3.6 and

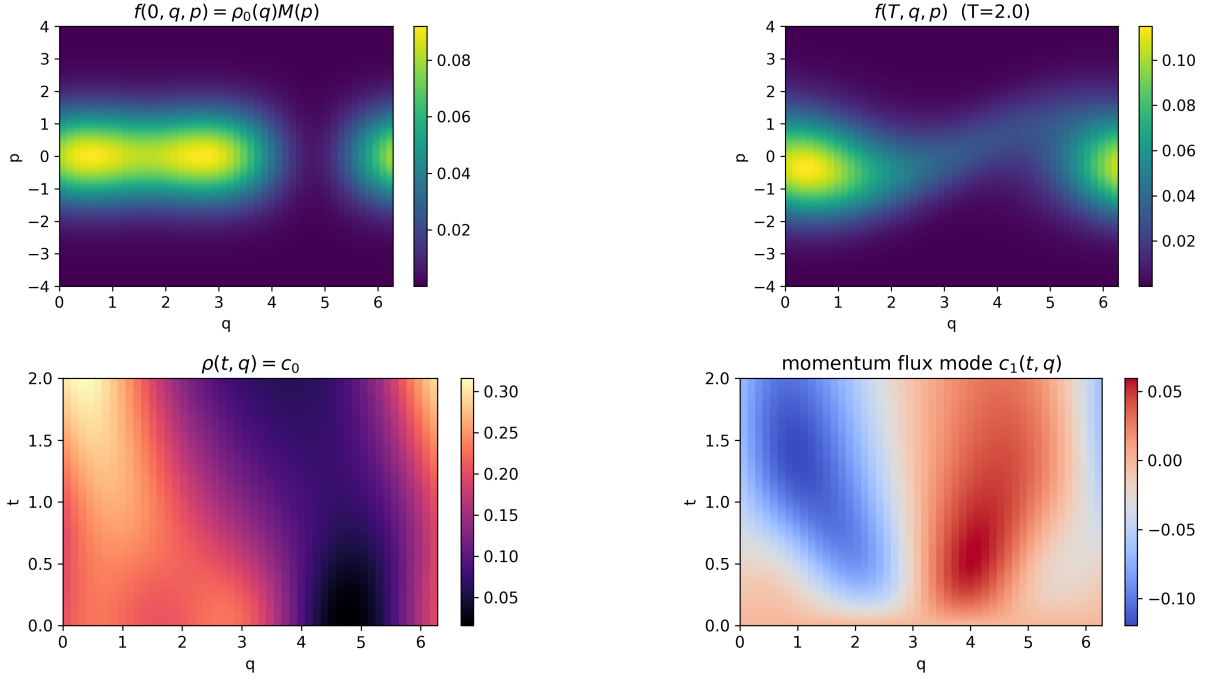


FIGURE 3. Top: phase-space density $f_{W_0}(t, q, p)$ at $t = 0$ (the product state $\rho_0 \otimes M$) and at $t = T = 2$, after streaming and the mean-field force have sheared it; measurement model ii. samples noisy point values of these surfaces at random (t_i, q_i, p_i) . Bottom left: the spatial density $\rho(t, q) = c_0(t, q)$ over the full time window. Bottom right: the first Hermite coefficient $c_1(t, q)$, i.e. the momentum flux. The force $F = -\partial_q(V + W * \rho)$ acts on c_1 , and higher levels, directly and on ρ only through the streaming coupling $\partial_t c_0 \propto -\beta \partial_q c_1$; the transient pattern of c_1 is the imprint of W_0 that both observation models must read, directly or indirectly.

the model comparison of Section 3.7 confirm that this single indirect channel is statistically effective, and that it is the only channel.

3.2. Synthetic data and the discrete measurement models.

3.2.1. *Ground truth and initial state.* We fix the symmetric, mean-zero ground-truth interaction

$$W_0(q) = -\cos q + \frac{1}{2} \cos 2q - \frac{1}{4} \cos 3q,$$

whose exact cosine coefficients $(a_1, a_2, a_3, a_4, \dots) = (-1, 1/2, -1/4, 0, \dots)$ will serve as the yardstick for recovery, and the initial condition in local momentum equilibrium

$$\phi(q, p) = \rho_0(q) M(p), \quad \rho_0(q) \propto 1 + 0.6 \sin q + 0.3 \cos 2q,$$

normalised to unit mass; in Hermite variables, $c_0(0, \cdot) = \rho_0$ and $c_n(0, \cdot) = 0$ for $n \geq 1$. The spatial asymmetry of ρ_0 (the $\sin q$ component) is deliberate: it places the system away from every stationary state of the dynamics, drives a nontrivial momentum flux c_1 , and thereby excites the ladder through which W becomes visible. Figure 3 displays the resulting dynamics: the phase-space state is sheared and mixed between $t = 0$ and $t = T$, the density $\rho(t, q) = c_0$ develops the structure that the observations will see, and the (unobserved) flux mode $c_1(t, q)$ shows the transient momentum currents that carry the imprint of the force.

3.2.2. *Avoiding the inverse crime.* The ground-truth trajectory is generated at a deliberately finer and different resolution ($N_H = 20$ Hermite modes, $\Delta t = 0.01$, snapshots stored every 0.02) than the solver used inside the likelihood ($N_H = 14$, $\Delta t = 0.02$), on the common spatial grid $N_q = 48$. Committing the “inverse crime” — generating and fitting data with the identical discretisation — would let the sampler exploit discretisation artefacts

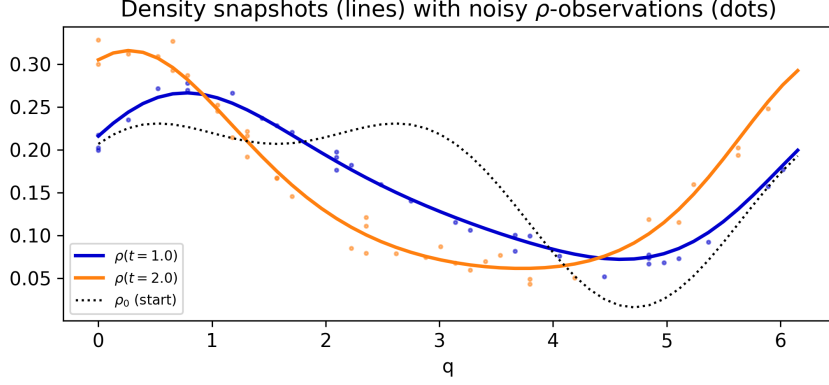


FIGURE 4. Density profiles at $t = T/2$ and $t = T$ (solid lines), the initial density (dotted), and the actual noisy observations $Y_i = \rho_{W_0}(t_i, q_i) + \epsilon_i$ falling on those snapshots (dots), at noise level $\sigma_\epsilon^\rho = 0.011$. Each of the 100 snapshot times carries on average 30 of the $N = 3000$ observations; the statistical problem is to deconvolve the interaction W from the way these profiles deform in time.

and produce artificially flattering reconstructions; with the present setup the model discrepancy between the two resolutions acts as a small additional, unmodelled error which the inference must absorb within the noise budget, as in any application to real data.

3.2.3. *Observation designs and noise.* From the fine trajectory we generate data sets of size $N = 3000$ for both measurement models:

- i. *macroscopic observations* $Y_i = \rho_{W_0}(t_i, q_i) + \epsilon_i$, with (t_i, q_i) drawn independently and uniformly from the product of the 100 positive snapshot times ($t = 0$ excluded, as $\rho(0, \cdot) = \rho_0$ is known) and the 48 spatial grid points, as a discrete counterpart of the design measure $\lambda = \lambda_T \otimes dq$ of the [Introduction](#);
- ii. *phase-space observations* $Y_i = f_{W_0}(t_i, q_i, p_i) + \epsilon_i$, with momenta drawn uniformly, $p_i \sim \text{unif}[-3.5\beta^{-1/2}, 3.5\beta^{-1/2}]$, i.e. within 3.5 thermal speeds; the point evaluations $f_{W_0}(t_i, q_i, p_i) = \sum_n c_n(t_i, q_i) e_n(p_i)$ are assembled through the precomputed matrix $B_{in} = e_n(p_i)$. The compactly supported uniform law stands in, at finite N , for a bounded design density ν concentrating on the informative low-momentum region, in the spirit of the Gaussian choice discussed in the [Introduction](#): beyond the cutoff, f_{W_0} , whose observed values range over $[0, 0.113]$, is smaller than the noise level and observations there would be essentially uninformative.

The noise is i.i.d. centred Gaussian with known standard deviation. In the phase-space experiment of Section 3.5.1, we take $\sigma_\epsilon = 0.004$, which is 15% of the empirical standard deviation 0.027 of the clean signal $\{f_{W_0}(t_i, q_i, p_i)\}_{i \leq N}$. In the model comparison of Section 3.7, the two observables must be compared fairly: a value of ρ (momentum-integrated, signal standard deviation 0.073) and a value of f (a single phase-space sample, signal s.d. 0.027) live on different scales, so fixing one common σ_ϵ would covertly favour one model. We therefore calibrate the two noise levels to a common noise-to-signal ratio,

$$\frac{\sigma_\epsilon^f}{\text{sd}(f_{W_0}(t_i, q_i, p_i))} = \frac{\sigma_\epsilon^\rho}{\text{sd}(\rho_{W_0}(t_i, q_i))} = 0.15 \quad \implies \quad \sigma_\epsilon^f = 0.0040, \quad \sigma_\epsilon^\rho = 0.0110,$$

so that any difference between the posteriors is attributable to the observation model, not to unequal effective noise. Figure 4 shows density snapshots with the actual noisy ρ -observations overlaid, giving a sense of the information content of a single time slice.

3.3. **Empirical forward stability.** Before any inference we test numerically the local Lipschitz continuity of the solution map established in Lemma 2.4 from Assumptions 2.2–2.3. We perturb the truth by random even

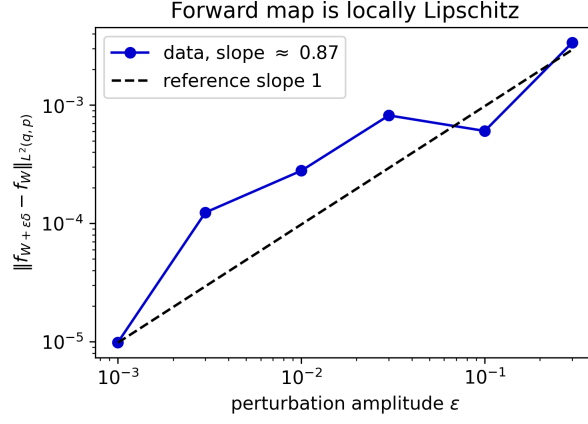


FIGURE 5. Empirical verification of the conditional Lipschitz estimate (Reg). Random even perturbations of amplitude ε are added to W_0 and the L^2 response of the phase-space state at $t = T$ is recorded; the log-log slope $0.87 \leq 1$ against the slope-1 reference (dashed) confirms a locally affine forward map with bounded differential over two and a half decades of perturbation size.

trigonometric polynomials

$$W_\varepsilon = W_0 + \varepsilon \delta W, \quad \delta W(q) = \sum_{m=1}^5 \frac{g_m}{m^2} \cos(mq), \quad g_m \sim_{\text{i.i.d.}} \mathcal{N}(0, 1),$$

over amplitudes spanning two and a half decades, $\varepsilon \in \{10^{-3}, 3 \times 10^{-3}, \dots, 3 \times 10^{-1}\}$, and record the response to such perturbation $\|f_{W_\varepsilon}(T) - f_{W_0}(T)\|_{L^2(\mathbb{T} \times \mathbb{R}, dq dp)}$ of the full phase-space state at the final time. On a log-log scale (Figure 5) the responses line up with fitted slope $0.87 \leq 1$ across the whole range: the change in the solution is controlled linearly by the change in the interaction, with bounded constant and no sign of instability, blow-up, or amplification of high-frequency perturbations: the perturbations here are weighted towards low frequencies precisely as the $H^{-\kappa}$ norm in (Reg) weights them. This is the behaviour asserted by (Reg): locally around W_0 the forward map is affine with bounded differential, which is the property through which Definition 1.1 enters the contraction theory of Section 1.

3.4. The prior. The prior is a band-limited Gaussian process prior of exactly the projected type described in Remark 5. On the span E_M of the cosine frequencies $1 \leq m \leq M$, we set

$$W(q) = \sum_{m=1}^M a_m \cos(mq), \quad a_m \sim \mathcal{N}(0, \tau^2 m^{-2s}) \text{ independent}, \quad (\Pi_M)$$

with $\tau = 1$, $s = 3/2$ in the baseline experiments and M varied below. By construction every draw is even and mean-zero, so the identifiability normalisation and the physical symmetry of (W) are hard-coded: the non-identifiable odd directions and the flat direction of constants are excised from the parameter space at the outset, as prescribed by the cosine-only construction after Condition 2.1. The RKHS of the untruncated series ($M = \infty$) is the even, mean-zero subspace of the Sobolev space $H^s(\mathbb{T})$ with norm $\|W\|_{\mathcal{H}}^2 = \tau^{-2} \sum_m m^{2s} a_m^2$, while sample paths, whose Sobolev norms have expectation $\mathbb{E}\|W\|_{H^t}^2 \asymp \sum_m m^{2t-2s}$, lie almost surely in $H^t(\mathbb{T})$ exactly for $t < s - 1/2$: the usual $1/2$ -derivative gap between a Gaussian prior and its RKHS [27, 28]. Figure 6 shows draws: smooth, even, mean-zero, and otherwise unconstrained shapes on the scale of the truth.

The decay exponent $s = 3/2$ is milder than the prescription of Condition 2.1, which at $d = 1$ requires RKHS regularity $H^{\alpha+1}$ with $\alpha + 1 > 5/2$; at any finite truncation M all draws are trigonometric polynomials, so the embedding requirements are trivially met and s acts purely as the regularising weight, but an asymptotic ($N \rightarrow \infty$) study would call for the stronger decay. Also, the N -dependent shrinkage of the rescaled prior (Π_N) is not enacted: at a single fixed sample size the rescaling amounts to a choice of the hyperparameter τ , and a

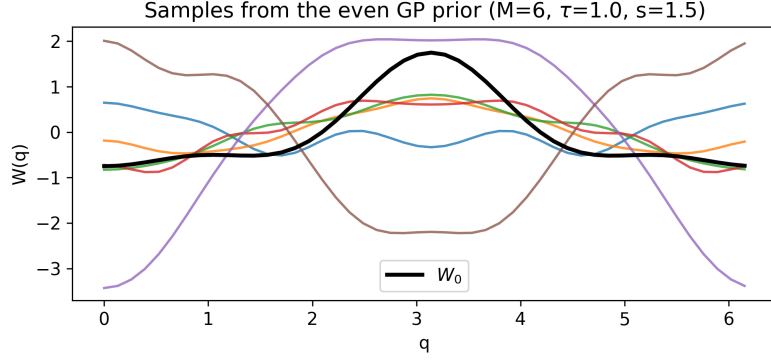


FIGURE 6. Six draws from the prior (Π_M) with $M = 6$, $\tau = 1$, $s = 3/2$ (thin colours), against the ground truth W_0 (thick black). Every draw is even, mean-zero and smooth, but otherwise unconstrained; the truth is of typical prior magnitude, consistent with the hypothesis $W_0 \in \mathcal{H}$ of Theorems 2.5–2.6.

numerical verification of the contraction rate δ_N , requiring ensembles of experiments across a ladder of sample sizes, is beyond our present scope. Neither simplification affects the finite-sample phenomena studied here, which concern the identifiability structure of the experiment rather than rates.

For sampling it is convenient to *whiten* the prior: writing $a_m = \tau m^{-s} \xi_m$, the coordinates $\xi = (\xi_m)_{m \leq M}$ are a priori standard Gaussian, $\xi \sim \mathcal{N}(0, I_M)$, and all algorithms below operate on ξ ; we write W_ξ for the potential with whitened coordinates ξ .

3.5. Likelihood and posterior sampling by pCN. Given whitened coordinates ξ , one forward solve at the inference resolution produces the coefficient trajectory $\{c_n(t_\ell, \cdot)\}$ on the snapshot grid, and the two log-likelihoods are assembled as

$$\ell_N^\rho(\xi) = -\frac{1}{2(\sigma_\epsilon^\rho)^2} \sum_{i=1}^N (Y_i - c_0(t_i, q_i))^2, \quad \ell_N^f(\xi) = -\frac{1}{2(\sigma_\epsilon^f)^2} \sum_{i=1}^N \left(Y_i - \sum_{n < N_H} c_n(t_i, q_i) B_{in} \right)^2,$$

with $B_{in} = e_n(p_i)$ precomputed once. Thus one likelihood evaluation costs one forward solve at the inference resolution, so the runtime of the sampler is that of the solver of Section 3.1 multiplied by the chain length.

The posterior is sampled by the preconditioned Crank–Nicolson algorithm of Section 1.2 in the whitened coordinates: from the current state ξ , propose

$$\xi' = \sqrt{1 - \beta_{\text{pCN}}^2} \xi + \beta_{\text{pCN}} \eta, \quad \eta \sim \mathcal{N}(0, I_M),$$

and accept with probability $1 \wedge \exp\{\ell_N(\xi') - \ell_N(\xi)\}$. The parametrisation corresponds to $\beta_{\text{pCN}} = \sqrt{2\delta}$ in the notation of Section 1.2; the proposal is the exact transition kernel of an Ornstein–Uhlenbeck process preserving the prior $\mathcal{N}(0, I_M)$, whence its reversibility with respect to the prior and the reduction of the Metropolis–Hastings ratio to the likelihood ratio alone [23]. Because this structure survives the limit $M \rightarrow \infty$, the algorithm is well defined on the function space and its mixing does not degenerate as the discretisation of the parameter is refined, a property that the truncation sweep of Section 3.6 exploits and confirms (the measured acceptance rates are essentially flat in M , from 25.6% at $M = 3$ to 28.2% at the fully parametric $M = 23$, at fixed step size).

In the baseline runs we use chains of 2500 iterations with the step tuned to acceptance rates between 16% and 25% ($\beta_{\text{pCN}} = 0.12$ giving 20.0% for model ii.; 0.09 giving 16.2% for model i.), each chain completing in 50–60 seconds; the first fifth of every chain is discarded as ‘burn-in’, and posterior summaries are computed from the remaining 2000 states.

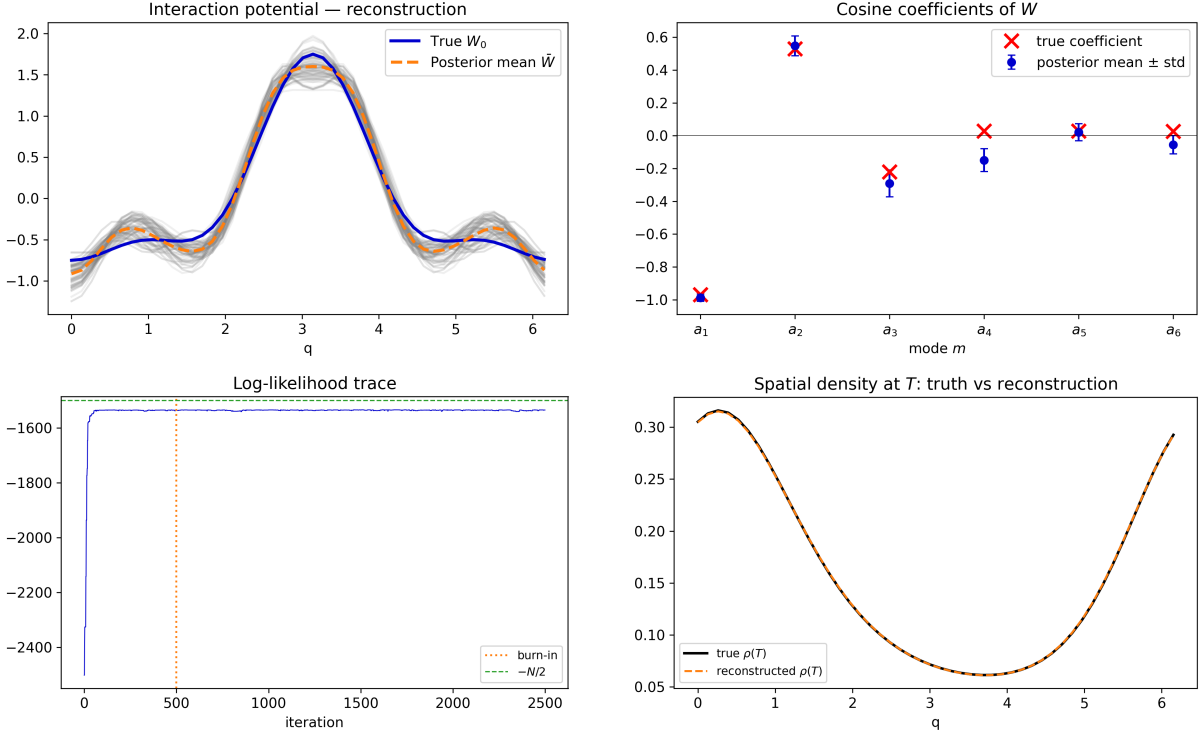


FIGURE 7. Posterior for measurement model ii. ($M = 6$). Top left: ground truth W_0 (solid), posterior mean $\hat{W}_N$ (dashed) and 80 posterior draws (grey), whose spread is the pointwise credible band. Top right: posterior mean $\pm$ one standard deviation of the cosine coefficients against the truth (crosses); a_1, a_2 are pinned, a_3 is recovered within about one standard deviation, and from a_4 on the posterior returns to the prior scale around zero. Bottom left: log-likelihood trace with burn-in marked; the chain equilibrates within a few hundred iterations onto the plateau $-N/2$ (dashed green), the value at which the mean squared residual equals the noise variance. Bottom right: the reconstructed density $\rho_{\hat{W}_N}(T, \cdot)$ overlays the truth to $\approx 10^{-3}$, four times below the noise level: the observable is fit essentially perfectly even where the parameter is only partially identified.

	a_1	a_2	a_3	a_4	a_5	a_6
truth	-1	+0.5	-0.25	0	0	0
posterior mean	-0.987	+0.548	-0.291	-0.149	+0.021	-0.055
posterior s.d.	0.022	0.060	0.082	0.070	0.052	0.055

TABLE 1. Cosine coefficients of W : exact ground truth against the pCN posterior for measurement model ii. ($M = 6$, $N = 3000$, $\sigma_\epsilon = 0.004$). The identified modes a_1, a_2, a_3 are recovered within about one posterior standard deviation; the unidentified a_4, a_5, a_6 carry prior-scale uncertainty, and their posterior locations are noise-driven; cf. discussion in Section 3.2.2.

3.5.1. *Posterior inference from phase-space observations.* Figure 7 and Table 1 summarise the posterior for measurement model ii. at the baseline configuration ($M = 6$, $\sigma_\epsilon = 0.004$, $N = 3000$).

Quantitatively: the identified coefficients land at $a_1 = -0.987 \pm 0.022$, $a_2 = 0.548 \pm 0.060$, $a_3 = -0.291 \pm 0.082$ against the exact values -1 , 0.5 , -0.25 , each within roughly one posterior standard deviation of the truth, and the overall reconstruction error is $\|\hat{W}_N - W_0\|_{L^2(\mathbb{T})} = 0.30$ on a potential of L^2 norm ≈ 2 . The unidentified coefficient $a_4 = -0.149 \pm 0.070$ exhibits a mild excursion of about two posterior standard deviations from its true value 0; as the analysis of Section 3.6 makes precise, the data carry essentially no information about this mode ($s_4 \approx 0$ below), so its posterior location is determined by the interaction of the noise realisation with the prior, and excursions on the prior scale are to be expected. Indeed, the matched run of Section 3.7, with an independent

noise realisation, places it at -0.104 ± 0.088 from density data. The log-likelihood trace equilibrates onto the plateau $-N/2 = -1500$ (measured plateau ≈ -1536), the value at which the empirical mean squared residual matches the noise variance: residuals fluctuating below the noise floor would signal overfitting, a plateau above it underfitting. And the reconstructed observable satisfies

$$\|\rho_{\hat{W}_N}(T, \cdot) - \rho_{W_0}(T, \cdot)\|_{L^2(\mathbb{T})} = 1.1 \times 10^{-3} \ll \sigma_\epsilon = 4 \times 10^{-3} :$$

the density is reproduced well below noise level although W itself is only partially identified.

3.6. Identifiability: the Jacobian ladder and reversion to the prior. The recovery pattern just described (low modes being pinned, high modes returning to the prior) is not an artefact of the sampler, and it can be predicted before any MCMC is run by linearising the forward map at the truth. This subsection carries out that analysis, verifies its exact agreement with the sampled posterior, and stress-tests the resulting picture in two directions: a ground truth with signal on a weakly identified mode, and the growth of the parameter dimension to the fully parametric limit.

We write $\mathbf{F}(\xi) = (f_{W_\xi}(t_i, q_i, p_i))_{i \leq N} \in \mathbb{R}^N$ for the prediction vector in whitened coordinates and let

$$J = \left. \frac{\partial \mathbf{F}}{\partial \xi} \right|_{\xi=\xi_0} \in \mathbb{R}^{N \times M}, \quad \xi_0 = \text{whitened coordinates of } W_0,$$

be its Jacobian, computed columnwise by central finite differences: $J_m \approx (\mathbf{F}(\xi_0 + h e_m) - \mathbf{F}(\xi_0 - h e_m))/2h$, $h = 10^{-3}$. Around the truth the log-likelihood is, to second order, the quadratic $-\|Y - \mathbf{F}(\xi_0) - J(\xi - \xi_0)\|^2/2\sigma_\epsilon^2$, so the singular value decomposition (SVD) $J = U\Sigma V^\top$, $\Sigma = \text{diag}(\varsigma_1 \geq \dots \geq \varsigma_M)$, diagonalises the local information geometry of the experiment: the log-likelihood curvature along the right-singular direction v_j is $(\varsigma_j/\sigma_\epsilon)^2$, to be compared with the whitened prior precision, which is 1 in every direction. A direction is *data-dominated* when $\varsigma_j \gg \sigma_\epsilon$ and *prior-dominated* when $\varsigma_j \lesssim \sigma_\epsilon$.

The right-singular vectors align with the individual cosine modes to three digits (dominant weights ± 1.00). The forward map does not mix the frequencies: this is an experimental confirmation of the diagonal action of the convolution $\nabla_q W * \rho$ in frequency space that underlies the deconvolution step (P) of Lemma 2.4. We may therefore identify singular direction j with cosine mode $m = j$. Moreover, the singular values decay geometrically, close to one order of magnitude per mode,

$$\varsigma_{1:6} = (1.73 \times 10^{-1}, 2.71 \times 10^{-2}, 1.96 \times 10^{-3}, 1.21 \times 10^{-4}, 1.16 \times 10^{-5}, 5.60 \times 10^{-7}),$$

i.e. ratios $\varsigma_m/\varsigma_1 = (1, 1.6 \times 10^{-1}, 1.1 \times 10^{-2}, 7.0 \times 10^{-4}, 6.7 \times 10^{-5}, 3.2 \times 10^{-6})$. This rapid decay is the empirical signature, for this experiment, of the statistical ill-posedness (smoothing) of the forward map that the theory encodes in the index κ of Lemma 2.4, here filtered through the specific excitation profile of ϕ and the finite horizon T : each additional frequency of W is roughly ten times harder to see. Now, looking at the per-experiment information $\iota_m = (\varsigma_m/\sigma_\epsilon)^2$ and comparing with the unit prior precision (Figure 8, left),

$$\iota_1 \approx 1.9 \times 10^3, \quad \iota_2 \approx 46, \quad \iota_3 \approx 0.24, \quad \iota_4 \approx 9 \times 10^{-4}, \quad \dots$$

the data-versus-prior crossover $\iota_m = 1$ falls between $m = 3$ and $m = 4$, precisely where the sampled posterior of Section 3.5.1 stopped tracking the truth.

The same object yields a fully Bayesian, mode-by-mode statement. Combining the local quadratic log-likelihood with the standard Gaussian prior on ξ gives the Laplace approximation of the posterior, a Gaussian with covariance

$$\Sigma = (I + \sigma_\epsilon^{-2} J^\top J)^{-1}$$

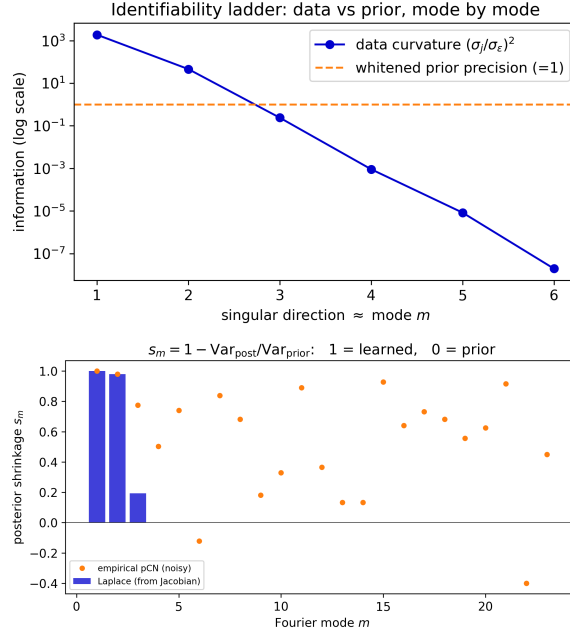


FIGURE 8. Top: per-mode data curvature $\iota_m = (\zeta_m/\sigma_\epsilon)^2$ from the SVD of the forward-map Jacobian at the truth (log scale), against the whitened prior precision 1 (dashed); the curvature falls by roughly an order of magnitude per mode and crosses the prior between $m = 3$ and $m = 4$, predicting the recovery ceiling of Figure 7 before any sampling. Bottom: Laplace posterior shrinkage $s_m = \iota_m/(1+\iota_m)$ (bars) across all 23 modes of the fully parametric prior, with the empirical shrinkage estimated from the pCN chain (dots). The two agree on the identified modes.

in whitened coordinates; the diagnostic we track is the *posterior shrinkage* of each mode,

$$s_m = 1 - \frac{\text{Var}_{\text{post}}(a_m)}{\text{Var}_{\text{prior}}(a_m)} = 1 - \Sigma_{mm} \in [0, 1],$$

in the spirit of the prior-versus-posterior comparisons of simulation-based inference: $s_m \rightarrow 1$ means the data have collapsed the prior on mode m ; $s_m \rightarrow 0$ means the posterior marginal is the prior marginal. Because the singular directions align with the coordinate modes, Σ is essentially diagonal and the shrinkage obeys the closed formula

$$s_m = \frac{\iota_m}{1 + \iota_m}, \quad \iota_m = \left(\frac{\zeta_m}{\sigma_\epsilon} \right)^2, \quad (\text{sh})$$

which the computed values verify exactly:

$$s_1 = 1.9 \times 10^3 / (1 + 1.9 \times 10^3) \approx 1.00, \quad s_2 = 46/47 \approx 0.98, \quad s_3 = 0.24/1.24 \approx 0.19, \quad s_4 \approx 9 \times 10^{-4} \approx 0.00,$$

matching the Laplace diagonal to the printed precision. Figure 9 displays the resulting prior-vs-posterior marginals for the first twelve modes at the fully parametric truncation ($M = 23$, below): modes 1 and 2 are narrow spikes displaced onto the true values; mode 3 sits on the crossover, partially contracted; from mode 4 upward the posterior curve lies exactly on the prior ($s_m = 0$). The bottom panel of Figure 8 overlays the empirical shrinkage estimated from the pCN chain: it agrees with the Laplace values on the informed modes but scatters wildly, including spuriously high and even negative values, on the uninformed ones.

3.6.1. A ground truth with energy on a weakly identified mode. The baseline truth W_0 has almost all of its energy in the identifiable modes, so its reconstruction tests little. A more demanding test places a coefficient of order one on a weakly identified mode:

$$W_{\text{hard}}(q) = 0.8 \cos q - 0.6 \cos 2q + 0.4 \cos 3q - 0.25 \cos 4q,$$

Prior vs posterior, mode by mode: the high modes fall back onto the prior

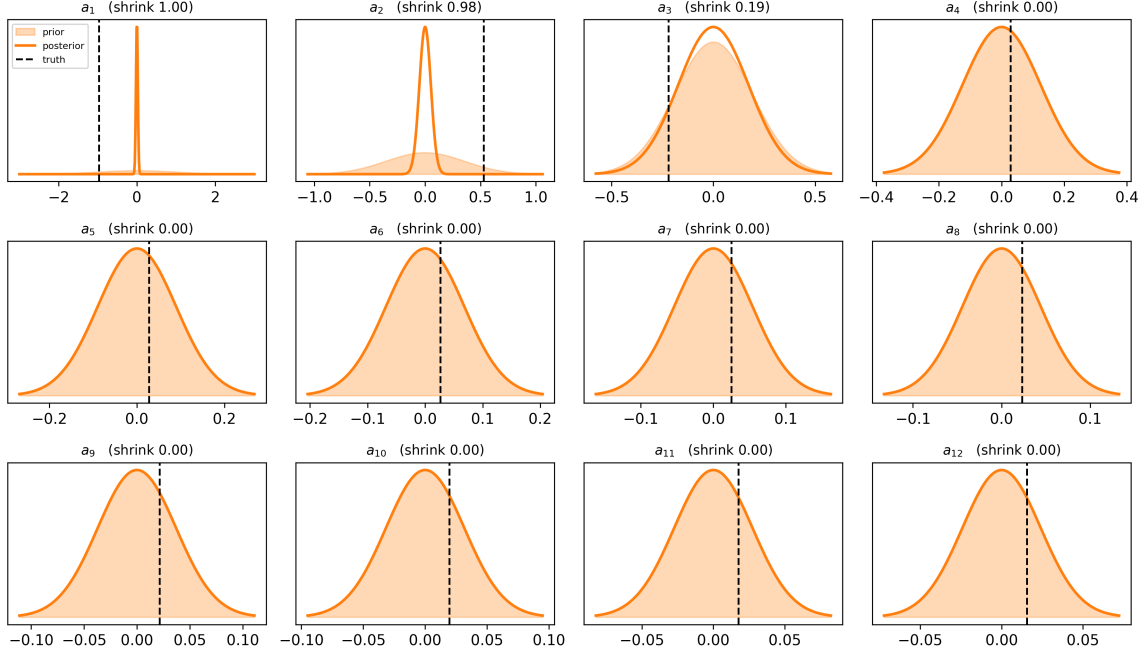


FIGURE 9. Prior (filled) versus Laplace posterior (thick lines) marginal for the coefficients $a_1, \dots, a_{12}$ at the fully parametric truncation $M = 23$, with the true values (dashed vertical lines) and the shrinkage s_m in each panel title. Modes 1–2: the posterior is a narrow spike centred on the truth ($s \approx 1$). Mode 3: partial contraction ($s \approx 0.19$). Modes ≥ 4 : the posterior density coincides with the prior density. This analysis was inspired by [47].

whose fourth coefficient sits, by the ladder just computed, at $t_4 \approx 10^{-3}$, three orders of magnitude below the prior. We regenerate the data from scratch with the fine solver, widen the prior to $M = 8$, and rerun the identical pCN (acceptance 39%, 3000 iterations). The outcome (Figure 10) is exactly the ladder’s prediction: $a_1 = 0.788 \pm 0.031$ and $a_2 = -0.582 \pm 0.034$ are recovered essentially exactly, $a_3 = 0.142 \pm 0.101$ captures only part of the true 0.4 and $a_4 = 0.010 \pm 0.039$ misses the true -0.25 entirely: the posterior for the loaded mode sits at zero, at prior scale, as if the signal were absent. The reconstruction $\hat{W}$ captures the gross two-well shape of W_{hard} and is blind to the $\cos 4q$ wiggle. We regard this negative result as an essential part of the evidence: the method recovers the low, dynamically excited modes of W and reverts to the prior, with prior-scale reported uncertainty, on the rest; testing only truths supported on the identifiable modes would have concealed this. Extending the reach of the experiment to higher modes would require a longer observation horizon, lower noise, stronger mean-field excitation, or an initial state ϕ engineered to pump those frequencies: the quantitative trade-offs that a stability estimate for the inverse problem, in the sense of Remark 5, would have to encode through the lower bounds on the Fourier coefficients of the excited density.

3.6.2. Growing the parameter dimension: the fully parametric limit. The truncation level M is a modelling choice, and a genuinely nonparametric method must tolerate its growth; on the N_q -point grid the largest resolvable cosine frequency is $N_q/2 - 1 = 23$, at which point the prior spans *every* spatial degree of freedom the discretisation carries. Sweeping $M \in \{3, 6, 12, 23\}$ with independent cold-started chains of 2000 iterations each, the reconstruction error is flat within its Monte Carlo variability,

$$\|\hat{W} - W_0\|_{L^2} = 0.197, 0.197, 0.222, 0.168 \quad \text{at} \quad M = 3, 6, 12, 23,$$

with the fully parametric run as accurate as any, and the four posterior means visually indistinguishable (Figure 11); the pCN acceptance rates are likewise flat (25.6%–28.2%), reflecting the discretisation-invariance of the

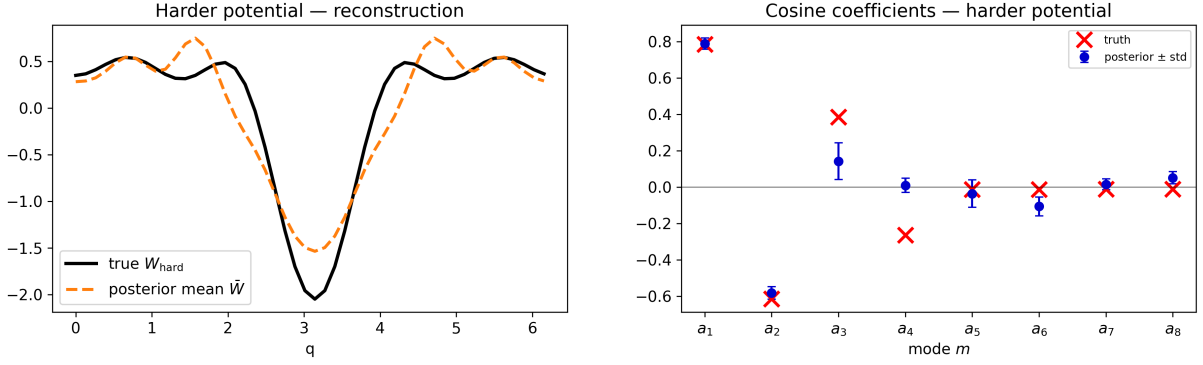


FIGURE 10. Stress test on W_{hard} , which loads the weakly identified fourth mode with the sizeable coefficient -0.25 . Left: the posterior mean captures the gross shape but not the $\cos 4q$ oscillation. Right: coefficient-wise, a_1, a_2 are exact, a_3 is partially recovered, and a_4 is missed entirely, as its posterior sits at zero with a prior-scale standard deviation. The failure is confined to the mode at which the identifiability ladder of Figure 8 places ι_m below the prior precision.

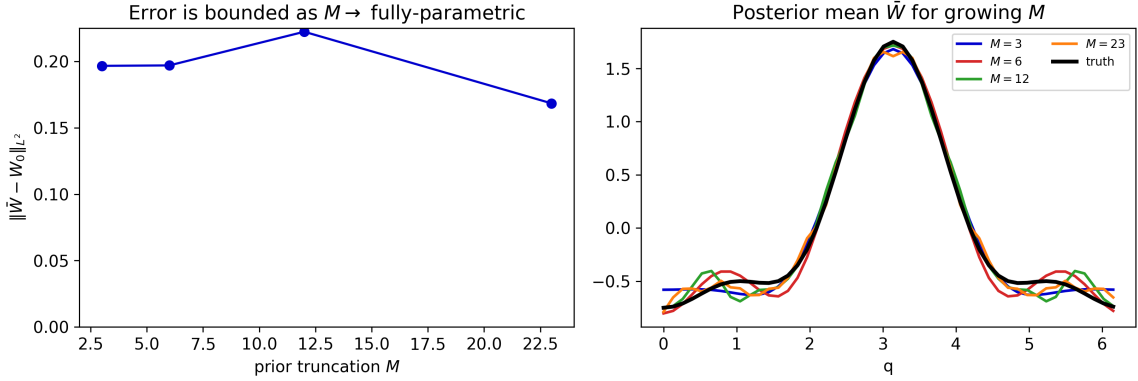


FIGURE 11. Robustness in the fully parametric limit (model ii.). Left: L^2 error of the posterior mean as the prior truncation grows from $M = 3$ to the fully parametric $M = 23$; the error stays in a narrow band with no degradation. Right: the posterior means for all four values of M overlaid on the truth; the extra 20 prior-dominated modes of the $M = 23$ run alter the reconstruction only within the width of the credible band.

algorithm. The mechanism is the summability of the prior variances: modes beyond the identifiable range have $\iota_m \approx 0$, so by (sh) each contributes its (small) prior variance $\tau^2 m^{-2s}$ to the posterior spread of W and nothing to its mean; the total excess variance $\sum_{m>3} \tau^2 m^{-2s}$ is bounded uniformly in M . An unpenalised least-squares fit of 23 coefficients to the same data would instead invert the near-singular normal matrix $J^\top J$ (condition number $\gtrsim (\zeta_1/\zeta_{23})^2$) and amplify noise catastrophically. This self-regularisation is the finite-sample manifestation of the role of the Gaussian prior in Theorem 1.2, where the shrunk prior controls exactly the complexity terms in the small-ball and entropy conditions, and it corroborates numerically the assertion of Remark 5 that the projected priors lose nothing as the truncation level grows through the resolvable range.

3.7. Density-only observations and the comparison of the two models. Observing the full phase-space state might be suspected of making the reconstruction artificially easy: model ii. hands the likelihood every momentum slice of f_{W_0} , an idealisation few experiments could match, whereas the macroscopic density of model i. is the physically natural observable and, as Remark 3 anticipates, the two models coincide at the level of forward contraction rates while potentially differing at the level of the inverse problem. We test the comparison head to head under the matched-noise protocol of Section 3.2: the two data sets are built on the same space-time design

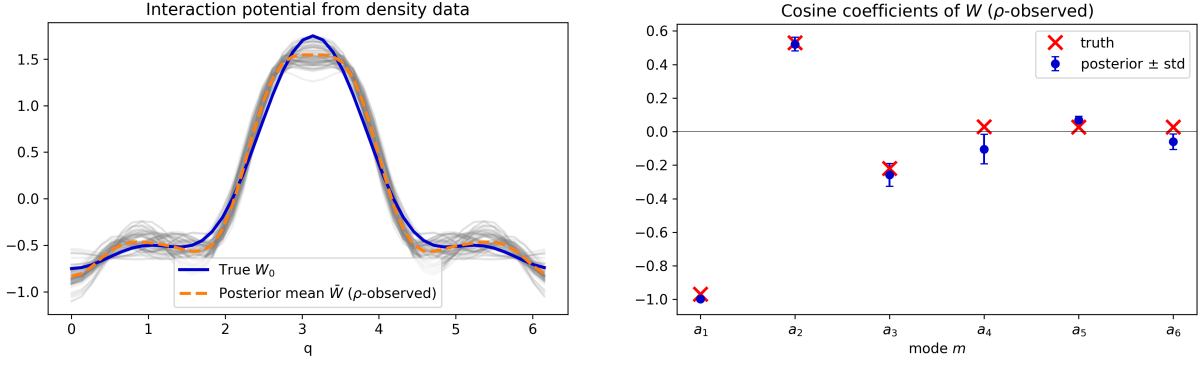


FIGURE 12. Posterior from density-only data (measurement model i., $M = 6$, $\sigma_\epsilon^\rho = 0.011$). Left: truth, posterior mean and 80 posterior draws. Right: coefficient recovery, cf. Figure 7.

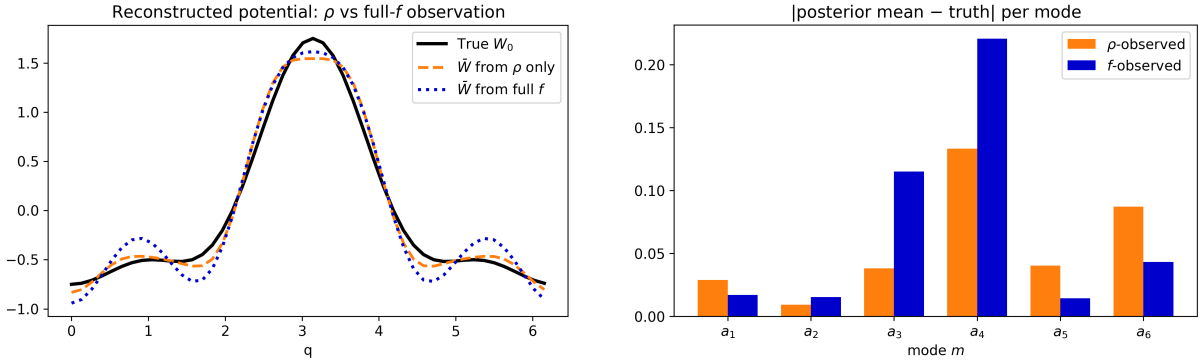


FIGURE 13. Head-to-head comparison at matched noise-to-signal ratio 0.15, common space-time design and identical prior and sampler. Left: posterior means from density-only data (dashed) and full-state data (dotted) against the truth. Right: coefficient-wise error of the posterior mean; neither observable wins systematically, and both fail together from mode 4 on.

$(t_i, q_i)_{i \leq N}$ from the same fine-solver trajectory with noise calibrated to the common noise-to-signal ratio 0.15, and the identical prior ($M = 6$) and pCN sampler are run on both likelihoods.

3.7.1. Recovery from the density alone. Despite discarding every bit of momentum information, the density data alone deliver the identified part of W (Figure 12): the posterior lands at

$$a_1 = -0.998 \pm 0.014, \quad a_2 = 0.521 \pm 0.040, \quad a_3 = -0.258 \pm 0.067$$

against the truth $(-1, 0.5, -0.25)$, each within one posterior standard deviation, and in fact *tighter* than the matched full-state posterior on the leading modes, with the higher modes reverting to the prior as before, at overall error $\|\hat{W} - W_0\|_{L^2} = 0.25$. The log-likelihood plateaus at $-N/2$ (measured ≈ -1489) and the reconstructed density at the final time satisfies $\|\rho_{\hat{W}}(T) - \rho_{W_0}(T)\|_{L^2} = 4 \times 10^{-4} \ll \sigma_\epsilon^\rho = 0.011$. The macroscopic measurement model, the direct kinetic analogue of the observation scheme of [1], and the model of Theorem 2.5, is thus fully adequate for the identifiable part of the interaction.

3.7.2. Head to head at matched noise. Figure 13 puts the two matched posteriors side by side. The reconstructed potentials overlay each other and the truth over the identified range; the per-mode errors $|\bar{a}_m - a_m|$ are comparable, with no systematic winner; and both posteriors stall at the same rung of the mode ladder. In this particular pair of noise realisations the density-based reconstruction is in fact the better one in L^2 (0.248 against 0.385), a pattern that recurs across seeds: a density value is a momentum-averaged functional of the state and hence a statistically cleaner read of the borderline mode than a single phase-space sample. At matched signal-to-noise,

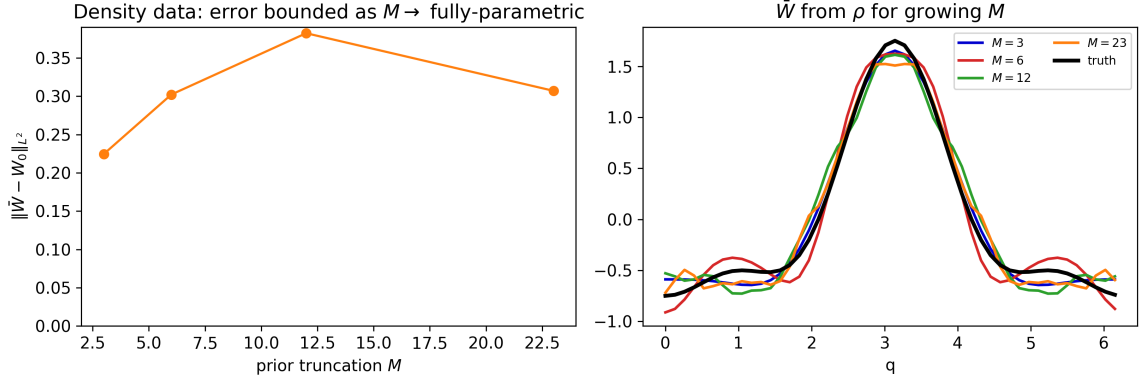


FIGURE 14. Truncation sweep for the density-only model. As for the full-state model (Figure 11), the L^2 error remains bounded up to the fully parametric truncation $M = 23$ and the reconstructions for all M agree within the credible band; the slightly higher and more variable error level reflects the larger matched noise σ_ϵ^ρ , not any instability of the nonparametric limit.

observing the entire phase-space state thus buys no meaningful information about W beyond what the macroscopic density already carries.

3.7.3. Robustness of the density model in the fully parametric limit. Finally, the truncation sweep of Section 3.6 is repeated on the density likelihood, where the concern is sharper: if growing the parameter count were to cause instability, one would expect it to appear first for the less informative observable. Across $M \in \{3, 6, 12, 23\}$, however, the error moves within the band 0.22–0.38 with stable acceptance rates (23.8%–26.8%), the fully parametric run landing at 0.31 (Figure 14). The band sits somewhat higher, and wanders somewhat more, than for full-state data (at the higher noise level σ_ϵ^ρ the borderline and unidentified modes are more prior-dominated, so each contributes slightly more posterior variance) but the qualitative verdict is identical: the Sobolev prior absorbs the unresolved modes for the density model exactly as for the full-state model, and the absorbed modes are, one by one, the same ones.

3.8. Summary and outlook. The experiments are consistent with the conditional theory of Section 2 and indicate which of its hypotheses remain to be established, and in what form.

1. *The forward problem is “under control”.* The Hermite–Fourier discretisation reproduces the exact OU friction spectrum, conserves mass to machine rounding, holds the self-consistent kinetic equilibrium stationary up to a residual with the exact $O(\Delta t^2)$ Strang scaling, and agrees, at Monte Carlo accuracy, with an independent simulation of the particle system (IPS) sharing no code.
2. *The mechanisms assumed by the theory are visible and quantifiable.* The spectra of the blocks L_k display the hypocoercive gap (0 at $k = 0$; γ at $|k| = 1$; saturation near 4.3γ) that a proof of Assumption 2.2 would have to produce in twisted norms; and the measured response of f_W to interaction perturbations is linear with bounded constant (log-log slope 0.87) over two and a half decades, as the conditional Lipschitz estimate (Reg) of Lemma 2.4 asserts.
3. *Posterior inference recovers the identified modes and no others.* pCN over band-limited Gaussian priors recovers the dynamically excited cosine modes of W within one posterior standard deviation from either observable, fits the observables to below noise level, and returns prior-scale uncertainty on the modes the experiment cannot see; the reach is dictated by a computable identifiability ladder that predicts the posterior before sampling, is confirmed by a truth with energy on a weakly identified mode, and is invariant under growth of the parameter dimension to the fully parametric limit.
4. *The two measurement models are informationally equivalent.* At matched signal-to-noise, the density-only and full-state posteriors are statistically indistinguishable. This is the inverse-problem counterpart of Remark 3,

and it suggests that the theoretical programme can be pursued for the physically natural density model i. with no loss expected relative to the full-state model. The numerics moreover indicate what such a stability estimate must quantify: the lower bounds, in terms of the excitation of ϕ , the horizon T and the noise level, on the Fourier-mode information ι_m whose crossover the experiments locate so sharply.

5. *Towards a stability estimate for the inverse problem.* The contraction of Theorems 2.5–2.6 concerns the forward quantities ρ_W, f_W only; transferring it to the parameter itself requires a stability estimate in the sense of Section 1, i.e. an inequality of the form $\|W - W_0\|_{L^2(\mathbb{T})} \lesssim \|\mathcal{G}(W) - \mathcal{G}(W_0)\|_{L^2}^\eta$ on the regularisation sets $\mathcal{W}_{M,N}$, which the present work does not prove. The numerics of Section 3, however, are promising in precisely this respect, and they indicate the form such an estimate must take. First, the right-singular vectors of the linearised forward maps align with the individual cosine modes to three digits: the linearisation is diagonal in frequency, exactly the structure that the deconvolution step (P) exploits and that a stability proof would invert. Second, on the dynamically excited frequency range, the singular values, though decaying geometrically (the finite-sample signature of the κ -smoothing), are strictly positive and stable across noise realisations, truncation levels and both observation models, so that on band-limited sets E_K spanned by the excited modes the inversion is well conditioned; together with the locally affine response of Section 3.3 (log-log slope 0.87), this is the behaviour a Lipschitz-stable inverse on E_K would produce. Third, the mode-wise information ι_m located by the identifiability ladder is the finite-sample counterpart of the lower bounds on the spatial Fourier coefficients of the excited density that a deconvolvability condition on ϕ – ruling out initialisation at a stationary, in particular spatially homogeneous, state – must encode: the stress test on W_{hard} shows that where these lower bounds fail ($\iota_4 \approx 0$) no stability can hold, and the posterior correctly reverts to the prior there. A rigorous stability estimate for band-limited projected priors, converting Theorems 2.5–2.6 into a contraction theorem for $\|W - W_0\|_{L^2(\mathbb{T})}$ in the spirit of Theorem 2 of [1], is deferred to future work.

APPENDIX

We gather additional evidence that Assumption 2.3 holds as stated, by reviewing the main result established in [41] and its proof.

Consider the following SDEs on $\mathbb{R}^{2d}$:

$$dX_t^{(i)} = Z_i(t, X_t^{(i)}) dt + \sigma_i(t, X_t^{(i)}) dB_t \quad \text{for } t \in [0, T], \quad i = 1, 2 \quad (\text{SDE})$$

with $T > 0$ and where

$$Z_i : [0, T] \times \mathbb{R}^{2d} \rightarrow \mathbb{R}^{2d} \quad \text{and} \quad \sigma_i : [0, T] \times \mathbb{R}^{2d} \rightarrow \mathbf{M}_{2d,d}(\mathbb{R})$$

are nice enough measurable maps such that each (SDE) is well-posed for $i = 1, 2$. In our case of interest, we have $X_t^{(i)} = (\bar{q}_t^{(i)}, \bar{p}_t^{(i)})$ that solve the pair of identical mean-field Langevin equations (??), i.e.

$$\begin{aligned} d\bar{q}_t^{(i)} &= \bar{p}_t^{(i)} dt, \quad \text{with} \quad X_0^{(i)} = (\bar{q}_0^{(i)}, \bar{p}_0^{(i)}) \sim \phi \\ d\bar{p}_t^{(i)} &= -\nabla_q V(\bar{q}_t^{(i)}) dt - [\nabla_q W_i * \rho_{W_i}(t)](\bar{q}_t^{(i)}) dt - \gamma \bar{p}_t^{(i)} dt + \sqrt{2\gamma\beta^{-1}} dB_t, \end{aligned}$$

for the same interaction potentials W_1, W_2 as in Assumption 2.3; namely,

$$Z_i(t, q, p) \equiv \begin{pmatrix} p \\ -\nabla_q V(q) - [\nabla_q W_i * \rho_{W_i}(t)](q) - \gamma p \end{pmatrix} \quad \text{and} \quad \sigma_1(t, q, p) = \sigma_2(t, q, p) \equiv \begin{pmatrix} \mathbf{0}_d \\ \sqrt{2\gamma\beta^{-1}} \mathbf{I}_d \end{pmatrix}.$$

This setting indeed accounts for degeneracies since the generators of the corresponding Markov semigroups

$$P_{s,t}^{(i)} f(x) \equiv \mathbb{E}[f(X_{s,t}^{i,x})] \quad \text{for } f \in \mathcal{B}_0(\mathbb{R}^{2d}),$$

where $X_{s,t}^{i,x}$ solves (SDE) for $t \in [s, T]$ with initial condition $X_{s,s}^{i,s} = x \in \mathbb{R}^{2d}$, are given by

$$\mathcal{L}_t^{(i)} \equiv \text{tr} [a_i(t, \cdot) \nabla^2] + Z_i(t, \cdot) \cdot \nabla$$

where $a_i \equiv \sigma_i \sigma_i^\top / 2$ may be degenerate.

The assumptions are the following:

- (1) for any $0 \leq s \leq t \leq T$, the Markov semigroup $P_{s,t}^{(2)}$ maps $C_0^2(\mathbb{R}^{2d})$ functions into $C_0^2(\mathbb{R}^{2d})$ functions, ensuring that the backward Kolmogorov equation

$$\partial_s P_{s,t}^{(2)} f = -\mathcal{L}_s^{(2)} P_{s,t}^{(2)} f, \quad s \in [0, t], \quad t \in (0, T]$$

holds for any $f \in C_0^2(\mathbb{R}^{2d})$;

- (2) the difference between the diffusion matrices $a_1 - a_2$ is differentiable and the divergence $\nabla \cdot [(a_1 - a_2)(t, \cdot) \nabla f]$ can be bounded by some function $H_{a_1 - a_2}(t)$.

In our case, assumption (2) is always trivially satisfied since $a_1 - a_2 \equiv 0$; assumption (1) is instead in particular satisfied when the coefficients are C_0^2 in the state variable, which in our setting requires the additional marginal regularity discussed in Remark 6 below.

Now, Theorem 2.1 yields

$$H(f_1(t) | f_2(t)) \leq \frac{\beta}{4\gamma} \int_0^t \|\nabla_q W_1 * \rho_1 - \nabla_q W_2 * \rho_2\|_{L^\infty(\mathbb{R}^{2d})}^2(s) ds.$$

To derive this entropy estimate, [41] establish the following intermediate result:

$$\begin{aligned} \frac{d}{ds} \mathbb{E} \left[\log P_{s,t}^{(2)} f(X_s^{(1)}) \right] &= \mathbb{E} \left[\left\langle [Z_1 - Z_2 - \nabla \cdot (a_1 - a_2)](s, \cdot), \nabla \log P_{s,t}^{(2)} f \right\rangle (X_s^{(1)}) - |a_2^{1/2}(s, \cdot) \nabla \log P_{s,t}^{(2)} f|^2 (X_s^{(1)}) \right] \\ &\quad + \mathbb{E} \left[\nabla \cdot [(a_1 - a_2)(s, X_s^{(1)}) \nabla \log P_{s,t}^{(2)} f(X_s^{(1)})] \right]. \end{aligned}$$

However, in our case the last term on the right hand side of this equality vanishes and we can bound:

$$\frac{d}{ds} \mathbb{E} \left[\log P_{s,t}^{(2)} f(X_s^{(1)}) \right] \leq \frac{1}{4} \mathbb{E} \left[|a_2^{-1/2} [Z_1 - Z_2](s, \cdot)|^2 (X_s^{(1)}) \right]$$

and integrating gives the desired estimate (H) posed on the entire Euclidean space, instead of the torus.

In fact, since $X_s^{(1)}$ has law $f_1(s)$, integrating the differential inequality over $s \in [0, t]$ yields

$$\mathbb{E} \left[\log f(X_t^{(1)}) \right] - \mathbb{E} \left[\log P_{0,t}^{(2)} f(X_0^{(1)}) \right] \leq \frac{\beta}{4\gamma} \int_0^t \int_{\mathbb{R}^d \times \mathbb{R}^d} \left| [\nabla_q W_1 * \rho_1 - \nabla_q W_2 * \rho_2](s, q) \right|^2 f_1(s, q, p) dq dp ds, \quad (\text{E})$$

whose right-hand side takes the role of the f_1 -weighted functional of (H).

The L^∞ form stated in Theorem 2.1 of [41] follows from (E) by bounding the inner integral by its supremum in q and using that $f_1(s)$ has unit mass; it is therefore strictly weaker than what the argument produces.

It remains to recognise the relative entropy on the left-hand side of (E): the first term

$$\mathbb{E} \left[\log f(X_t^{(1)}) \right] = \int_{\mathbb{R}^{2d}} f_1(t) \log f$$

and for the second, since $X_0^{(1)} \sim \phi$ and

$$P_{0,t}^{(2)} f(x) = \int_{\mathbb{R}^{2d}} f(y) p^{(2)}(0, x; t, y) dy,$$

with $p^{(2)}$ the transition density of the second diffusion, then

$$\mathbb{E} \left[P_{0,t}^{(2)} f(X_0^{(1)}) \right] = \int_{\mathbb{R}^{2d}} \int_{\mathbb{R}^{2d}} \phi(x) p^{(2)}(0, x; t, y) f(y) dy dx = \int_{\mathbb{R}^{2d}} f_2(t) f,$$

because both diffusions are started from the same initial law ϕ ; Jensen's inequality then gives

$$\mathbb{E} \left[\log P_{0,t}^{(2)} f(X_0^{(1)}) \right] \leq \log \int_{\mathbb{R}^{2d}} f_2(t) f.$$

Substituting into (E),

$$\int_{\mathbb{R}^{2d}} f_1(t) \log f - \log \int_{\mathbb{R}^{2d}} f_2(t) f \leq \frac{\beta}{4\gamma} \int_0^t \int_{\mathbb{R}^d \times \mathbb{R}^d} \left| [\nabla_q W_1 * \rho_1 - \nabla_q W_2 * \rho_2](s, q) \right|^2 f_1(s, q, p) dq dp ds$$

and taking the supremum over $f = e^g$ with g bounded and measurable – after a routine approximation of such f by perturbed functions in C_0^2 to which the semigroup identity applies – the Gibbs/Donsker–Varadhan variational formula [48, Lemma 1.4.3]

$$H(f_1(t) | f_2(t)) = \sup_{g \in C_0(\mathbb{R}^{2d})} \left\{ \int_{\mathbb{R}^{2d}} f_1(t) g - \log \int_{\mathbb{R}^{2d}} f_2(t) e^g \right\}$$

identifies the left-hand side with the relative entropy.

This proves the equivalent of (H) on the phase space $\mathbb{R}^d \times \mathbb{R}^d$.

To pass from $\mathbb{R}^d \times \mathbb{R}^d$ to $\mathbb{T}^d \times \mathbb{R}^d$, we need a lifting argument that uses only the periodicity of the coefficients.

Write $\pi : \mathbb{R}^d \rightarrow \mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d$ for the covering projection, and regard the drifts

$$F_i(t, q) \equiv -\nabla_q V(q) - [\nabla_q W_i * \rho_i(t)](q), \quad i = 1, 2,$$

as given, time-dependent vector fields on $\mathbb{T}^d$, hence as $\mathbb{Z}^d$ -periodic vector fields on $\mathbb{R}^d$; here ρ_i denotes the spatial marginal of the *torus* solution f_i furnished by Assumption 2.2 i). Let ν be any probability density on $\mathbb{R}^d \times \mathbb{R}^d$ pushed forward to ϕ by $\pi \otimes \text{id}$, i.e.

$$(\pi \otimes \text{id})_* \nu = \phi,$$

for instance ϕ restricted to the fundamental domain $[0, 1]^d \times \mathbb{R}^d$ and extended by zero, and let $\tilde{X}^{(i)} = (\tilde{q}^{(i)}, \tilde{p}^{(i)})$ solve (SDE) on $\mathbb{R}^{2d}$ with drift $Z_i = (p, F_i(t, q) - \gamma p)^\top$ and initial law ν ; the drift is globally Lipschitz in the state variable, since $V \in C^\infty(\mathbb{T}^d)$ and convolution with the probability density $\rho_i(t)$ does not increase the Lipschitz

constant of $\nabla_q W_i$, so this Euclidean SDE is well-posed. Denote by $\tilde{f}_i(t)$ the law of $\tilde{X}_t^{(i)}$. By periodicity of f_i , the projected process $(\pi(\tilde{q}_t^{(i)}), \tilde{p}_t^{(i)})_{t \leq T}$ is a weak solution of the torus SDE with the same drift F_i and initial law ϕ , so its time- t law, namely $(\pi \otimes \text{id})_{\#} \tilde{f}_i(t)$, solves the *linear* Fokker–Planck equation on $\mathbb{T}^d \times \mathbb{R}^d$ with coefficients F_i and initial datum ϕ ; by uniqueness for this linear equation, which is contained in Assumption 2.2 i) once the nonlinear term is frozen, this law is $f_i(t)$. Two observations conclude the argument.

(a) *The left-hand side decreases under projection.* Relative entropy is contractive under measurable maps (the data-processing inequality), whence

$$H(f_1(t) | f_2(t)) \leq H(\tilde{f}_1(t) | \tilde{f}_2(t)).$$

(b) *The right-hand side is invariant under projection.* The integrand $|Z_1 - Z_2|^2(s, \cdot)$ is $\mathbb{Z}^d$ -periodic in q and independent of p , and $f_1(s)$ is the image of $\tilde{f}_1(s)$ under $\pi \otimes \text{id}$; therefore

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} |Z_1 - Z_2|^2(s, \tilde{q}) \tilde{f}_1(s, \tilde{q}, p) d\tilde{q} dp = \int_{\mathbb{T}^d \times \mathbb{R}^d} |Z_1 - Z_2|^2(s, q) f_1(s, q, p) dq dp.$$

Applying the Euclidean estimate to $(\tilde{X}^{(1)}, \tilde{X}^{(2)})$ and combining (a) and (b) yields (H) on $\mathbb{T}^d \times \mathbb{R}^d$, with the same constant $\beta/(4\gamma)$ and the f_1 -weighted right-hand side, which is Assumption 2.3 in the form used in Lemma 2.4. Note that the argument never uses the nonlinear structure of (VFP): the marginals ρ_i are frozen into the coefficients from the outset, exactly as in the Girsanov heuristic of Section 2, and the nonlinearity re-enters only in Step 1 of the proof of Lemma 2.4, where $\rho_1 - \rho_2$ is controlled by the entropy itself through Pinsker's inequality and the Grönwall loop is closed.

REMARK 6 (ON THE REGULARITY ACTUALLY REQUIRED). The one hypothesis of [41] that the present setting does not verify for free is assumption (1): that the frozen semigroup $P_{s,t}^{(2)}$ maps $C_0^2(\mathbb{R}^{2d})$ into itself, so that $P_{s,t}^{(2)} f$ solves the backward Kolmogorov equation classically and the identity for $\partial_s \mathbb{E}[\log P_{s,t}^{(2)} f(X_s^{(1)})]$ can be differentiated under the expectation. This is standard when the coefficients are of class C_0^2 in the state variable and continuous in time.

For the drift Z_2 the state-regularity requirement reads $\nabla_q W_2 * \rho_2(t) \in C_0^2(\mathbb{T}^d)$, i.e. three bounded derivatives of $W_2 * \rho_2(t)$; since the class (W) only provides $W_2 \in C^2$, one derivative must be borrowed from the marginal, via $\partial_q^3(W_2 * \rho_2) = \partial_q^2 W_2 * \partial_q \rho_2$, which is bounded as soon as $\rho_2(t) \in W^{1,1}(\mathbb{T}^d)$; Assumption 2.2 iv) supplies

$$\rho_2 \in L^2(0, T; H^{\kappa+1}(\mathbb{T}^d)) \hookrightarrow L^2(0, T; W^{1,1}(\mathbb{T}^d)),$$

i.e. this regularity for a.e. t and in a time-integrated sense only, not the uniform-in-time continuity that a direct application of [41] presupposes. The missing step is thus to run the identity above with the weaker, time-integrated regularity of the coefficients – for instance by mollifying ρ_2 in time, applying the estimate to the mollified drift, and passing to the limit using the stability of relative entropy under weak convergence, i.e. its lower semicontinuity, on the left and dominated convergence on the right – or alternatively to upgrade Assumption 2.2 iv) to $\rho_W \in C([0, T]; W^{1,\infty}(\mathbb{T}^d))$ uniformly over $\{\|W\|_{W^{2,\infty}} \leq M\}$, which the hypocoercive propagation-of-regularity estimates discussed after Assumption 2.2 are expected to deliver for $\kappa + 1 > d/2 + 1$. Either route is a matter of analysis of (VFP) rather than of the entropy argument itself, which is why we have stated (H) as an assumption; its verification is deferred to future work together with that of Assumption 2.2.

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